

An AI-generated landscape painting featuring a vast wetland or marsh area. In the foreground, tall, golden-brown grasses are in sharp focus. Beyond them, a body of water reflects the sky, surrounded by green marshland and distant blue hills. The sky is filled with numerous birds in flight, rendered in a painterly style with soft, blended colors. The overall mood is serene and naturalistic.

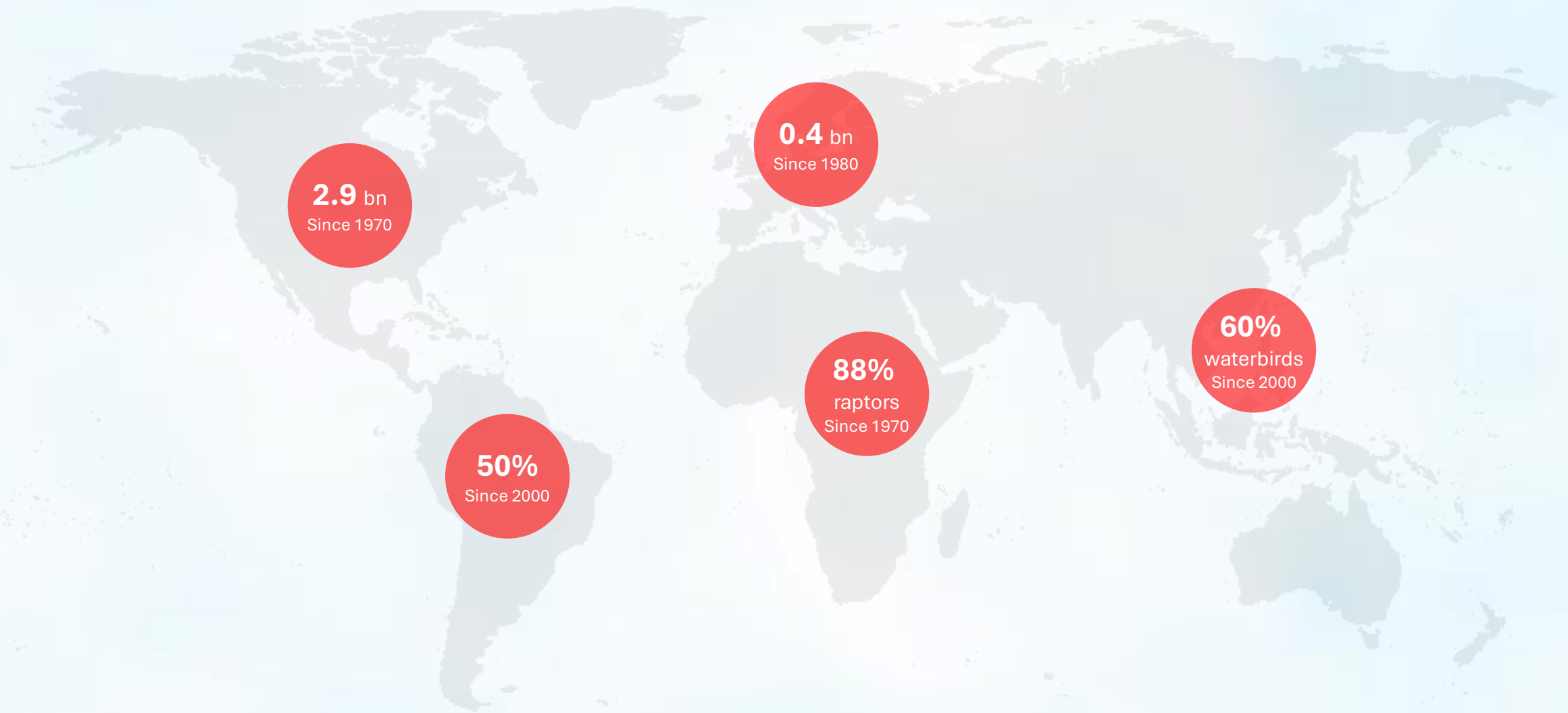
Spatiotemporal dynamics of North American breeding bird populations



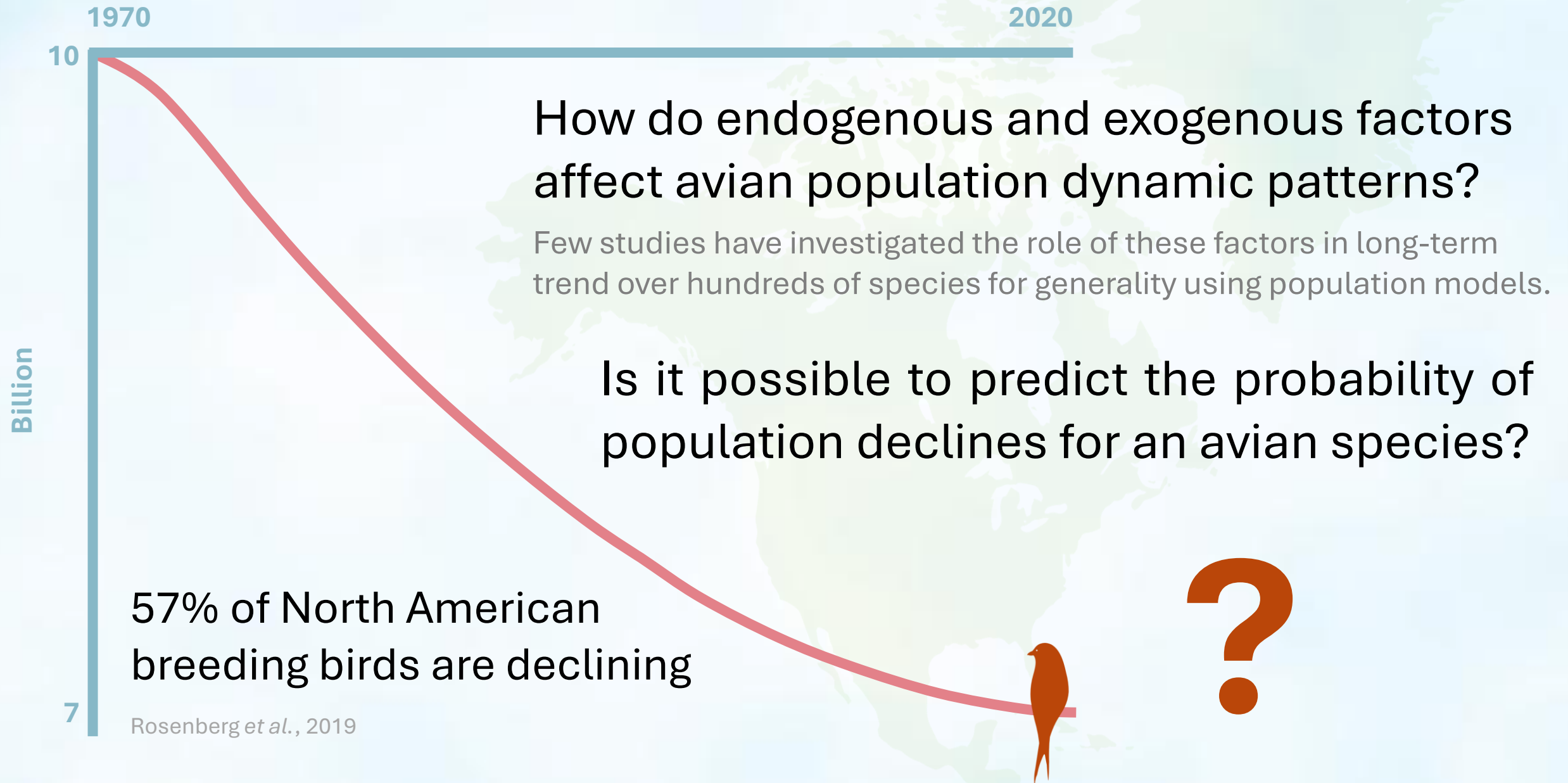
MISSISSIPPI STATE UNIVERSITY™
DEPARTMENT OF WILDLIFE,
FISHERIES AND AQUACULTURE

Wentao Song

Global declines of avian populations and biodiversity



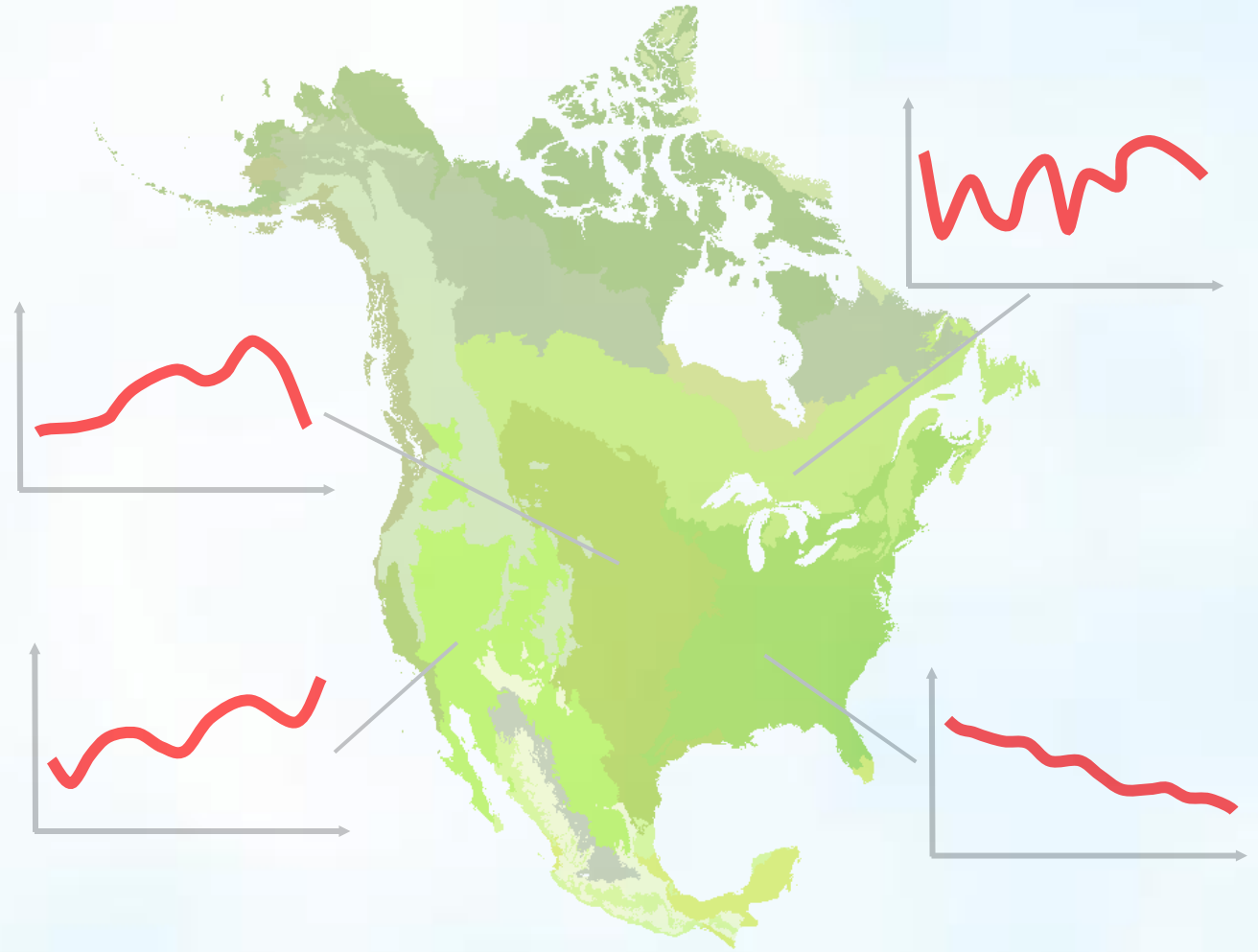
Rosenberg *et al.*, 2019; Rigal *et al.*, 2023; Blake *et al.*, 2024; Shaw *et al.*, 2024; BirdLife International, 2017



Overall trend
of a “species” population



Spatial variations
within a species



Objectives

1

Identify different patterns in the population trajectories of North American breeding birds for the past 50 years using machine learning;

2

Explain the identified dynamic patterns using life history theory, phylogeny, population regulation, and **global** climate change;

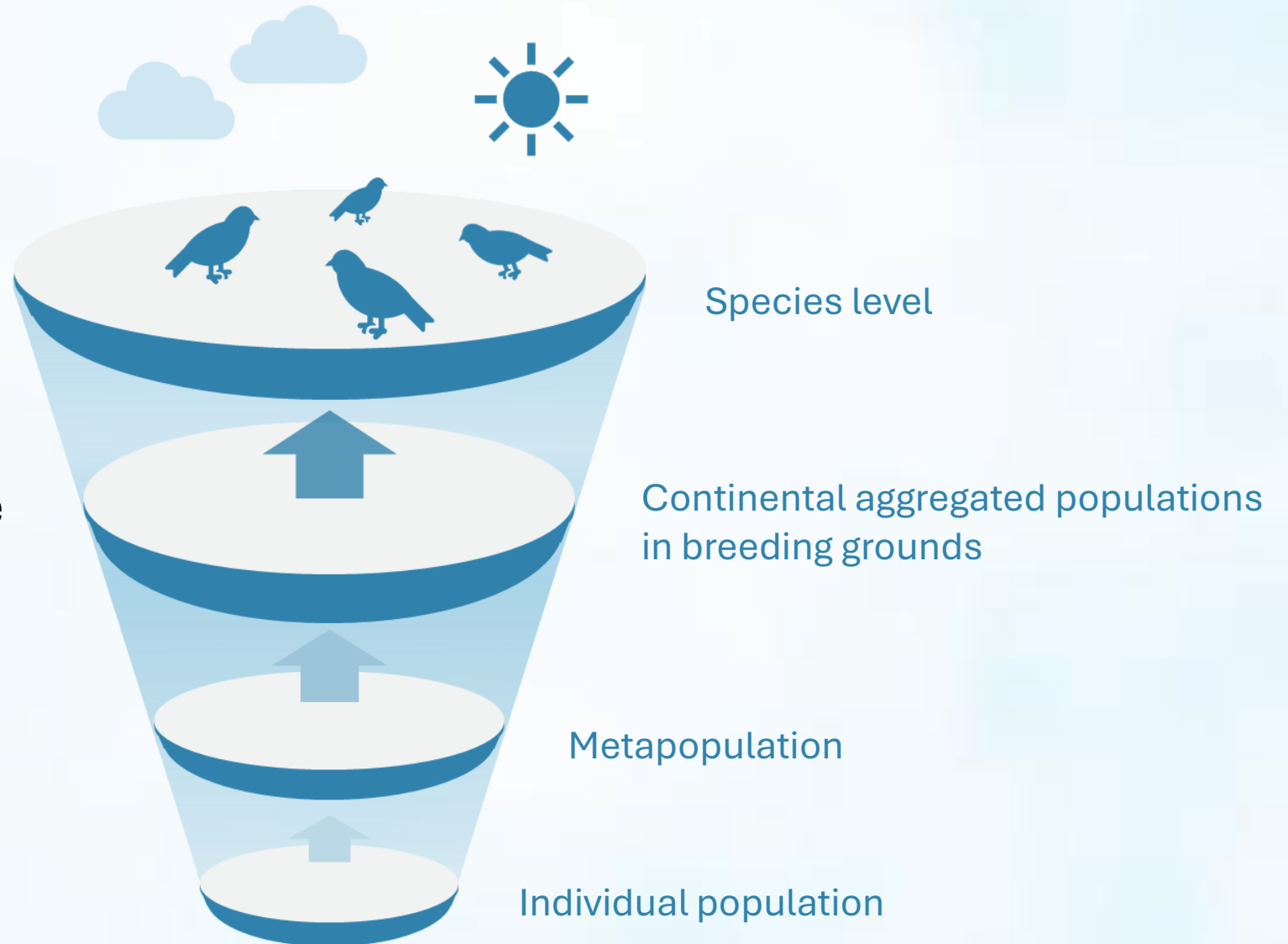
3

Investigate the spatial variations in avian population responses to **local** climate changes and avian abundances.

Organization scale of study

Population

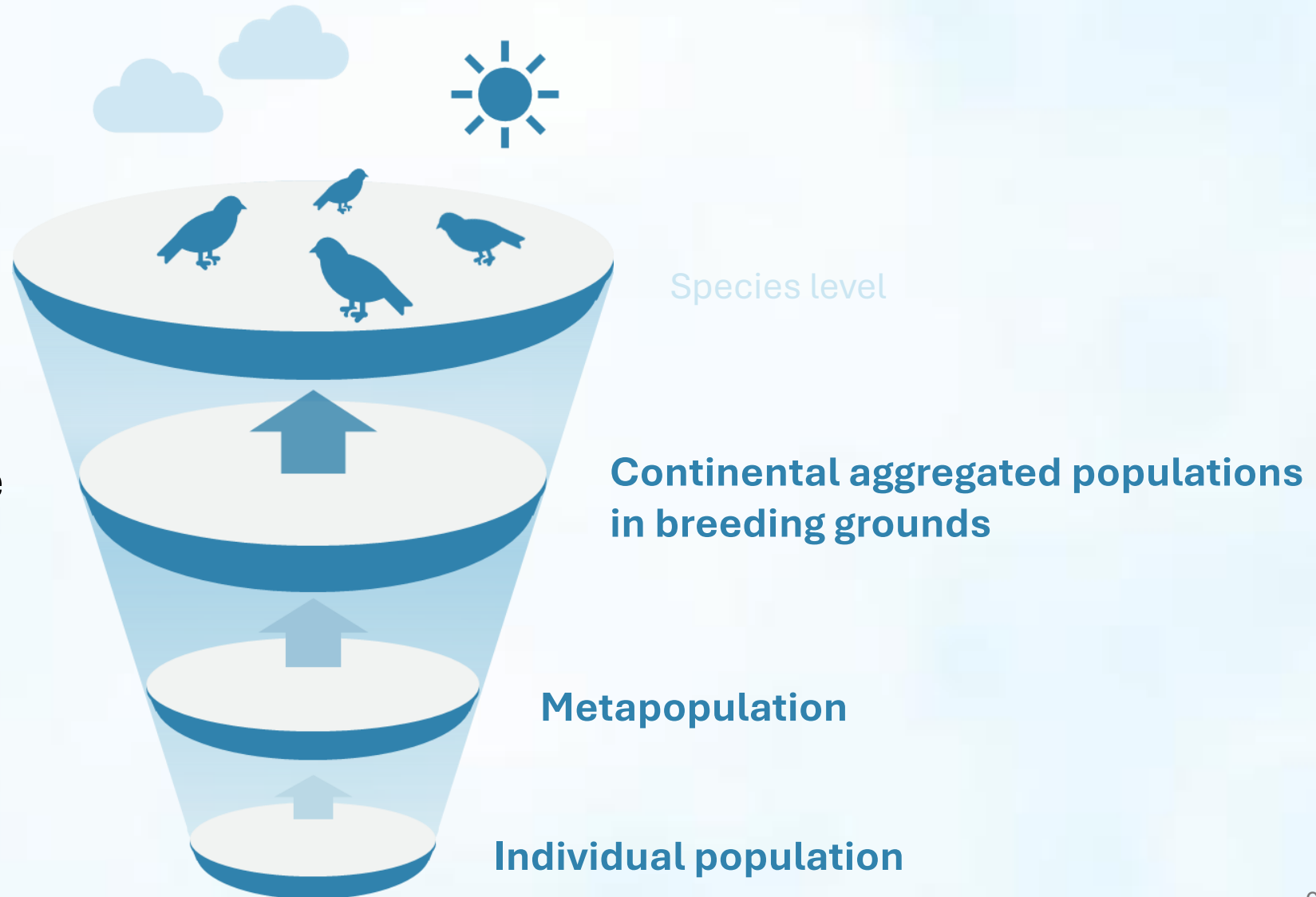
all the individuals of the same species that live in a particular area and a given time.



Organization scale of study

Population

all the individuals of the same species that live in a particular area and a given time.



Chapter 2

Long-term trends of North American Breeding Bird Populations

Objectives

1

Identify different patterns in the population trajectories of North American breeding birds for the past 50 years using machine learning;

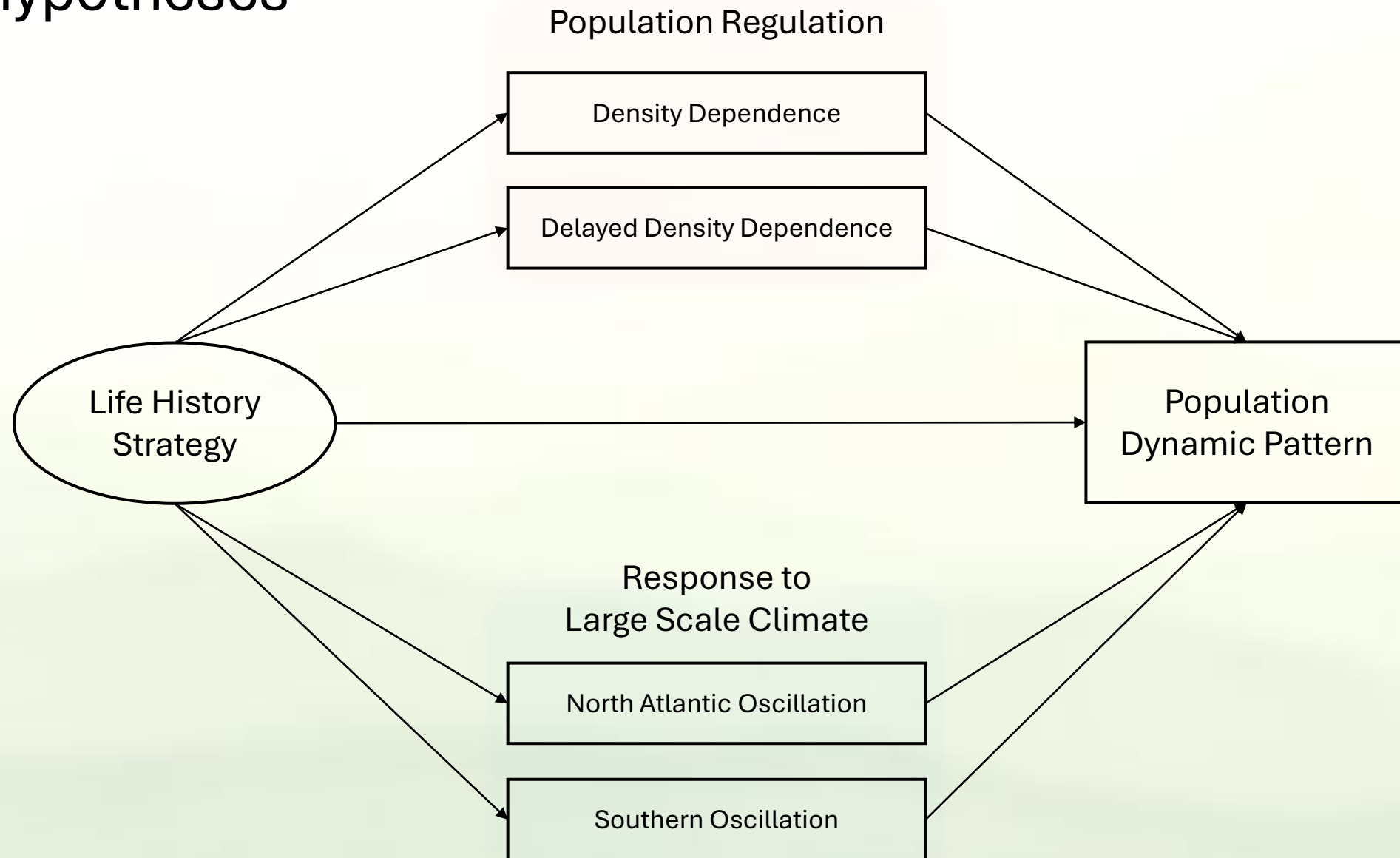
2

Explain the identified dynamic patterns using life history theory, phylogeny, population regulation, and **global** climate change;

3

Investigate the spatial variations in avian population responses to **local** climate changes and avian abundances.

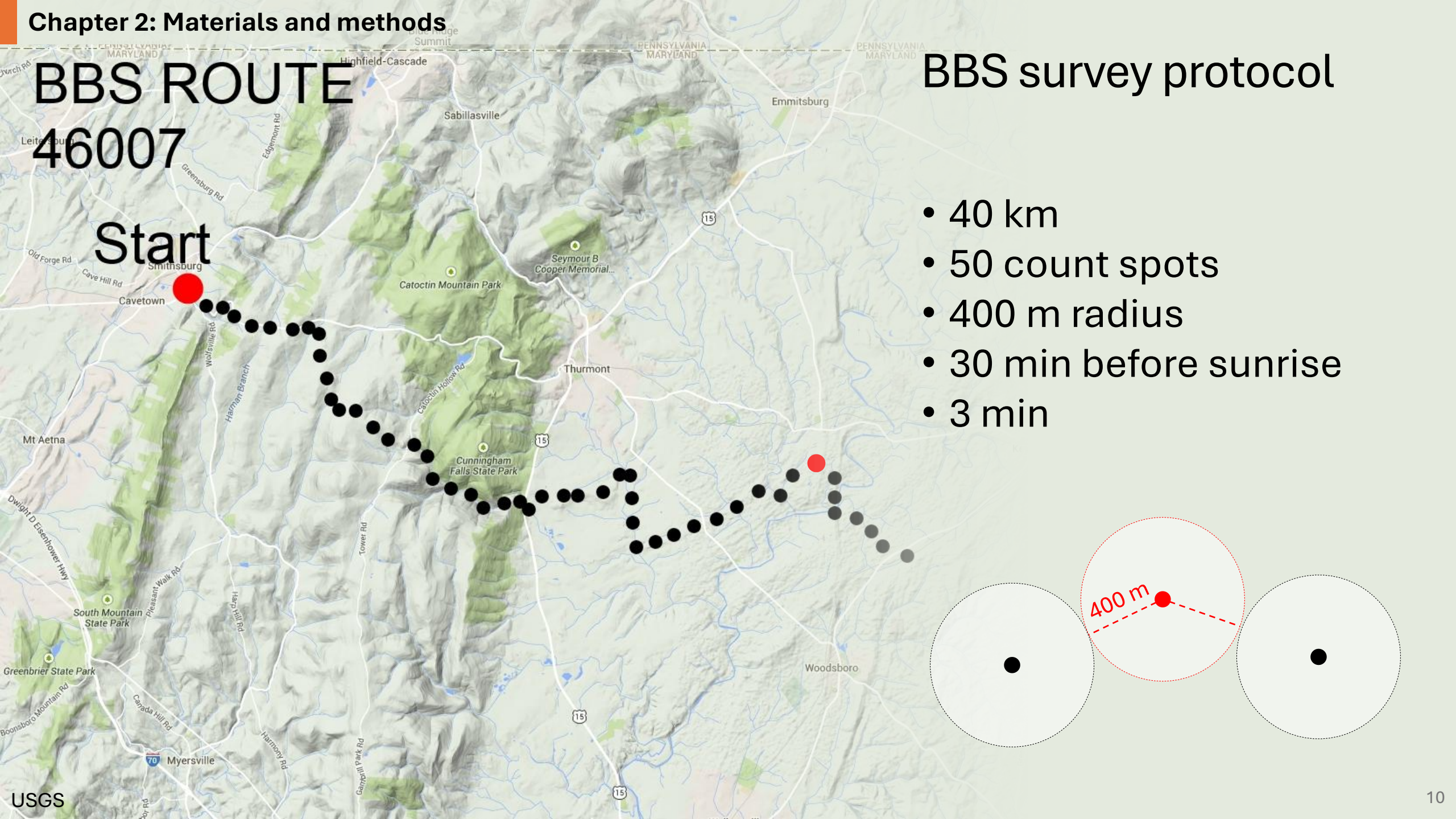
Causal hypotheses

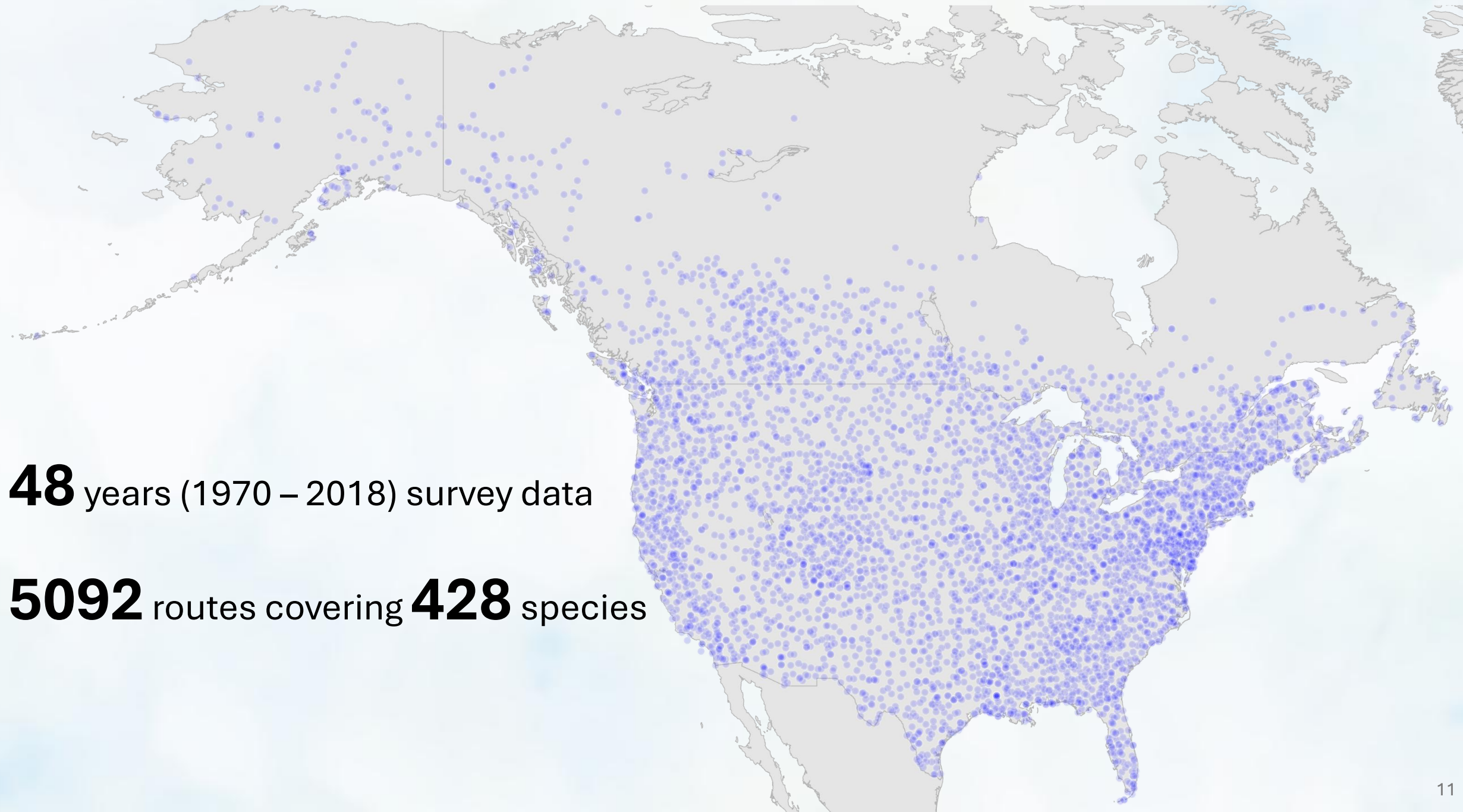


BBS ROUTE 46007

BBS survey protocol

- 40 km
- 50 count spots
- 400 m radius
- 30 min before sunrise
- 3 min



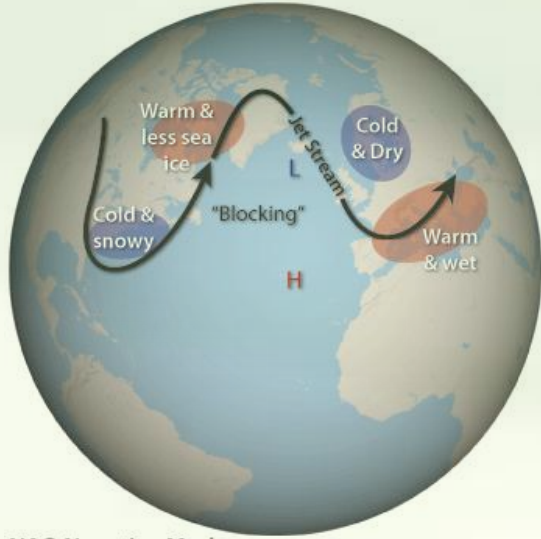


48 years (1970 – 2018) survey data

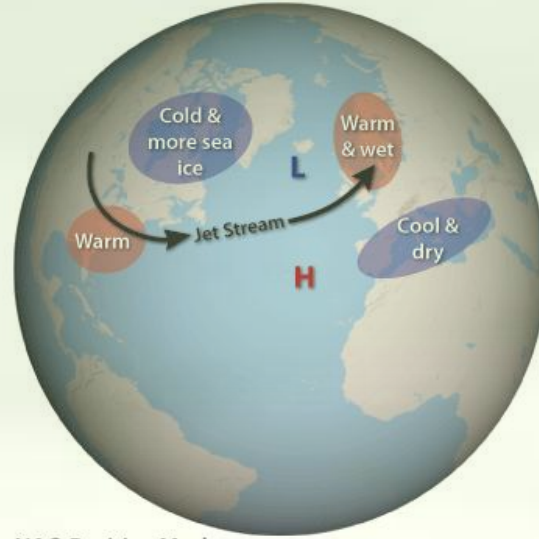
5092 routes covering **428** species

Life history traits

The amniote life-history database (Myhrvold <i>et al.</i> , 2015)	Avian hand-wing index (Sheard <i>et al.</i> , 2020)
Body mass The body mass of an adult individual in grams.	Range size Number of 1° x 1° grid cells overlapping breeding/resident range.
Longevity The lifespan of an individual in years.	Migration BirdLife International migration score. Four levels: altitudinal migrant, full migrant, nomadic, and not a migrant.
Annual clutches The number of clutches produced per year.	Territoriality Territoriality scores from Tobias <i>et al.</i> , 2016. Three levels: none, weak (seasonal), and strong (year-round).
Clutch size The size of the clutch.	Diet Eight diet categories: plants, nectar, seeds, fruit, invertebrates, vertebrates, scavenger, and omnivore.
Incubation The time between when the egg is laid and when it hatches in days.	Habitat Habitat scores from Tobias <i>et al.</i> , 2016. Three levels: dense, semi-open, and open.
Fledging age The time it takes for the bird to be capable of flight (or for some species, leaving the nest) in days.	



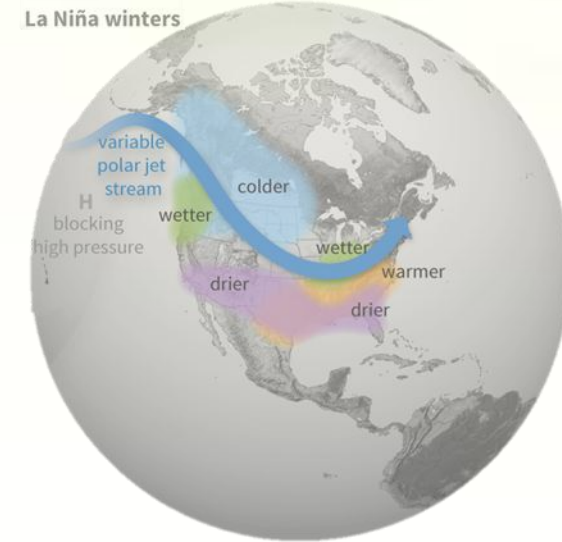
NAO Negative Mode



NAO Positive Mode



El Niño winters



La Niña winters

NAOI

The North Atlantic Oscillation index is based on the surface sea-level pressure difference between the Subtropical (Azores) High and the Subpolar Low.

SOI

Southern Oscillation index has been calculated based on the differences in air pressure anomaly between Tahiti and Darwin, Australia.

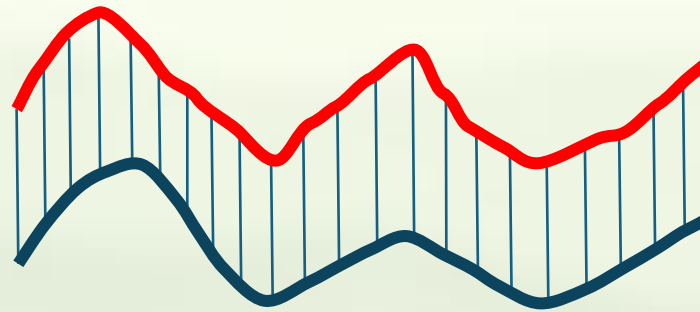
Previous methods for trend analysis

Sauer et al. 2011 $\text{Log}(\lambda_{s,j,t}) = S_i + \beta_i(t - t^*) + \omega_j + \gamma_{i,t} + \varepsilon_{s,j,t}$

Link et al. 2020 $\gamma_s(t) = \alpha_s + \beta_s(t - t^*) + \varepsilon_{st}$

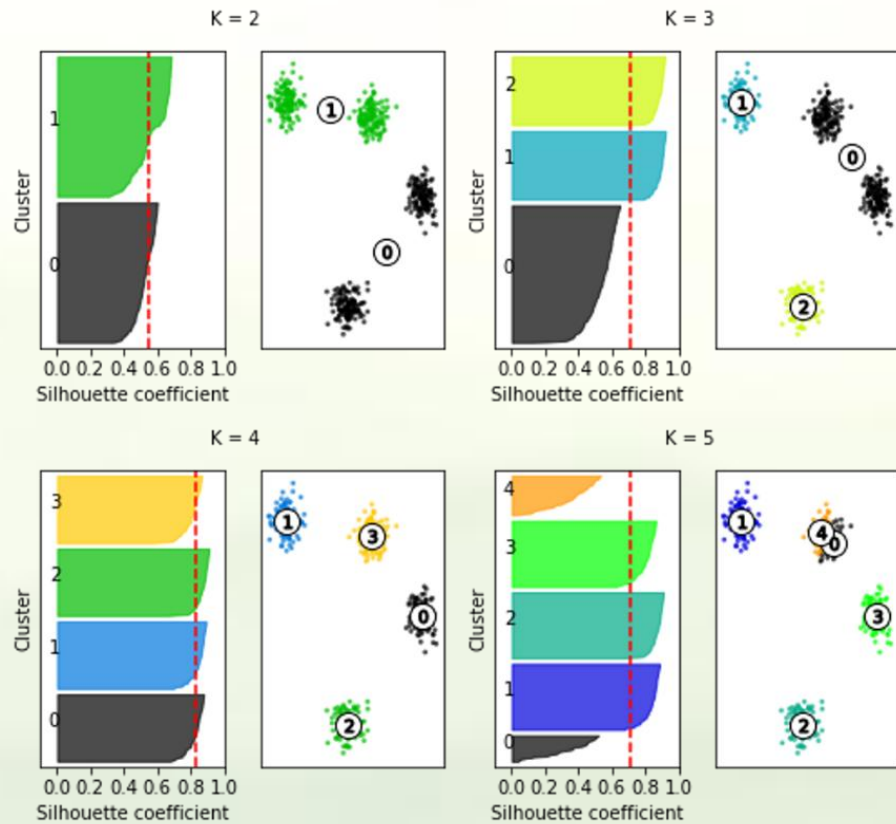
Smith et al. 2021 $\text{Log}(\lambda_{s,j,t}) = \theta_s + \Delta_s(t) + \eta/[j,t] + \omega_j + \varepsilon_{s,j,t}$

**Shape matching
in this study**

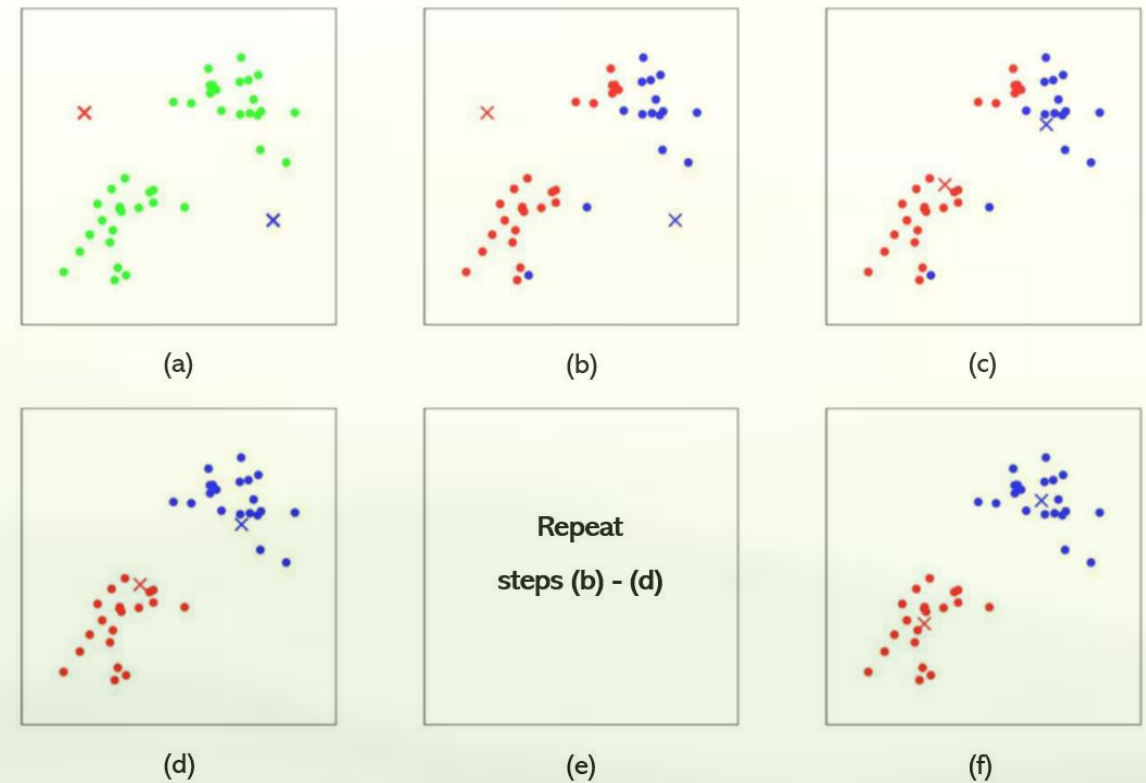


K-means clustering

Silhouette analysis



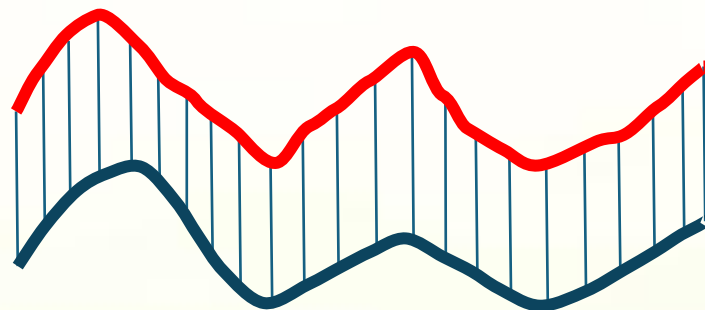
K-means clustering



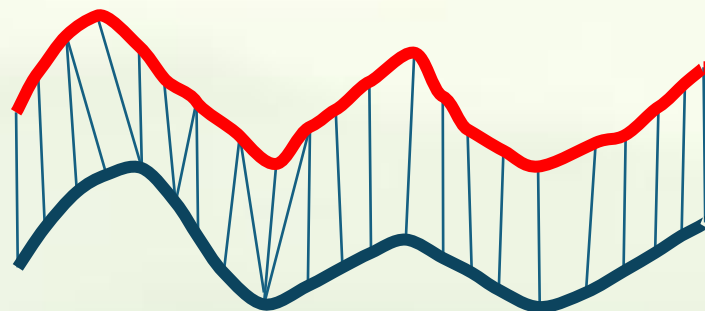
after Ng, 2010

Dynamic Time Warping (DTW)

Euclidean matching



Dynamic time warping



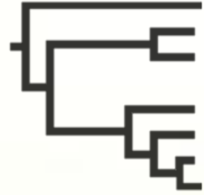
6	4	6
2	1	2
	3	1

8	14	16	11	14	10	8
5	9	11	7	10	7	11
3	7	8	6	7	8	12
5	6	8	5	8	6	10
6	4	6	4	9	6	9
2	1	2	4	5	8	15
	3	1	4	1	5	9

$$D_{i,j} = |x_i - y_j| + \min(D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1})$$

$$D_{2,2} = |1 - 6| + \min(4, 2, 1) = 6$$

Statistical models



Bayesian phylogenetic regression

for testing differences in the features among trend groups



State space models

for extracting features of population dynamics



Structural equation models

for testing hypotheses regarding direct and indirect effects of density dependence, density independence, and life history traits

Bayesian phylogenetic regression

Bayesian logistic regression with phylogenetic tree

trend $\sim$ life history traits + (1|gr(TipLabel, cov = A))

BirdTree.org



Gompertz state space population growth models

$$Y_t = X_t + u_t$$

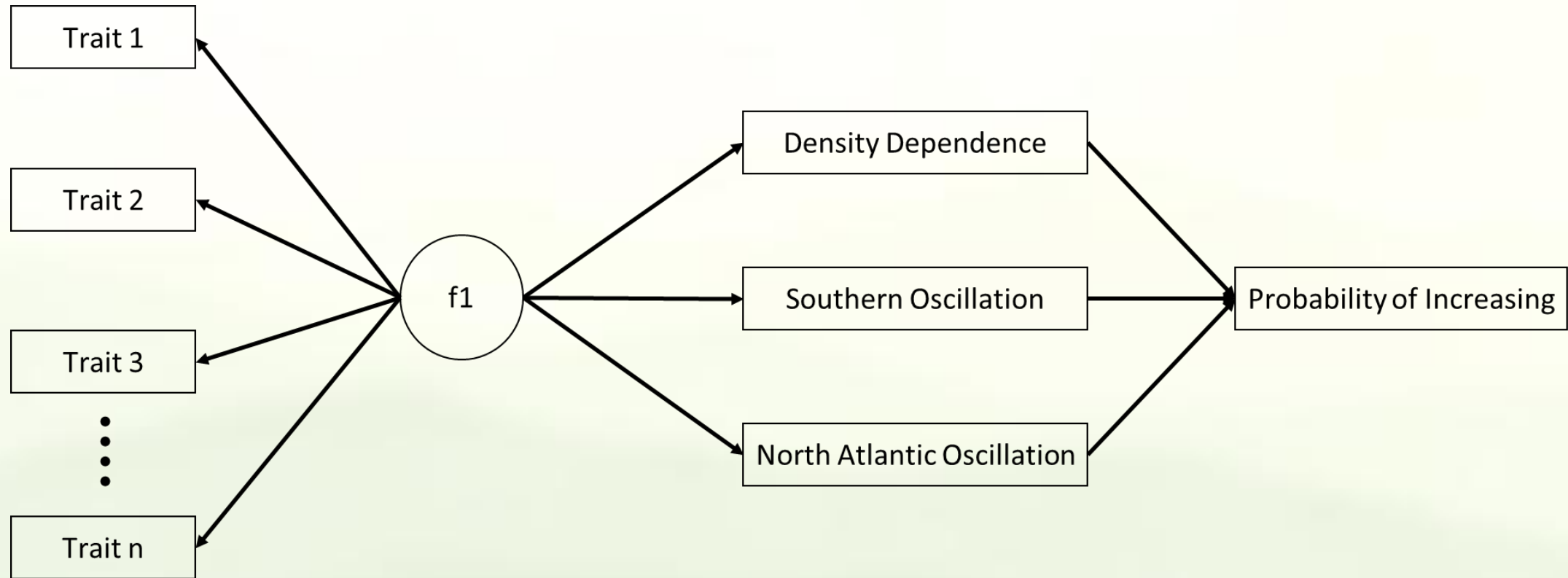
$$X_t = a + bX_{t-1} + cX_{t-2} + dNAOI_{t-1} + fSOI_{t-1} + e_t$$

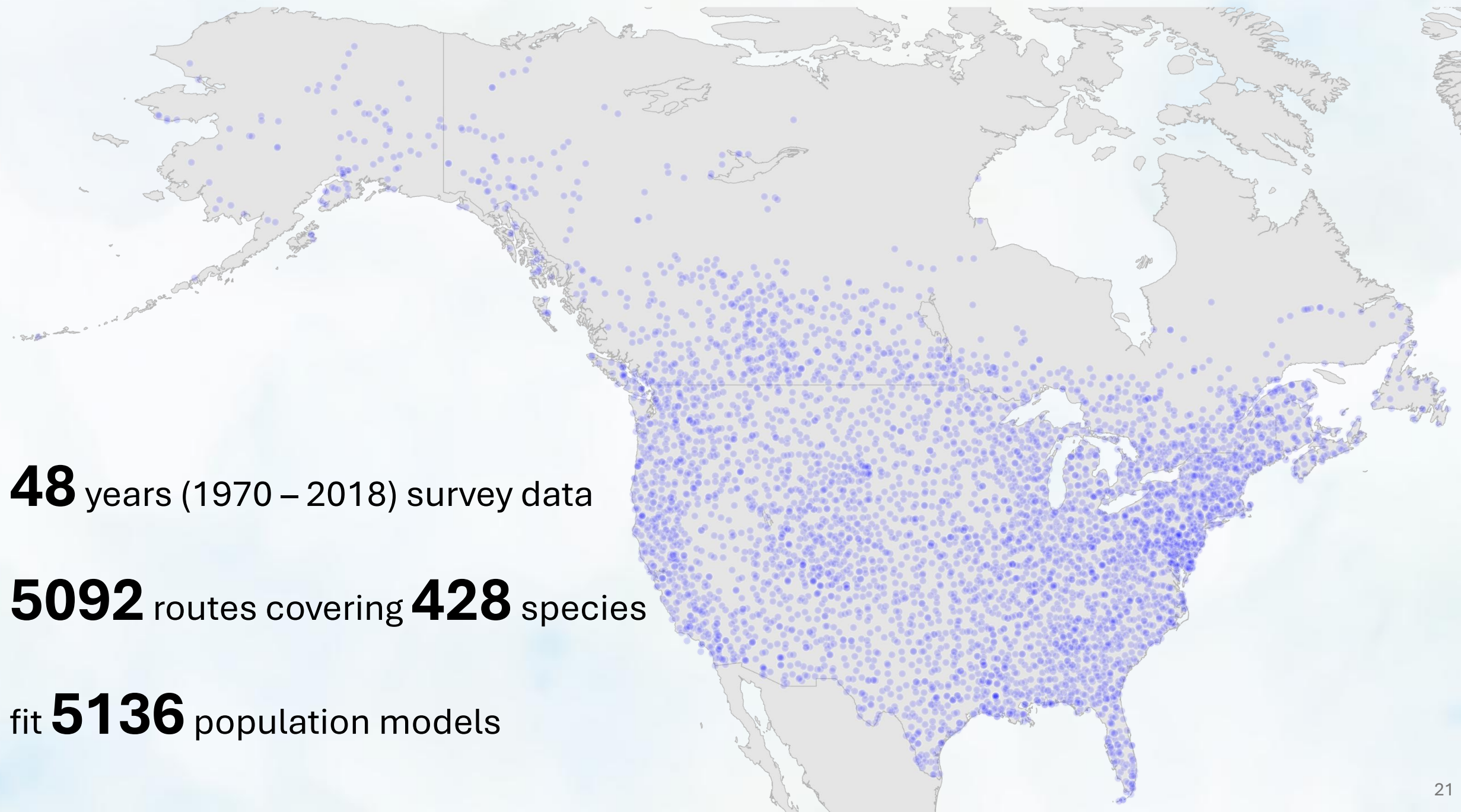
- b density dependence
- c delayed density dependence
- d North Atlantic Oscillation
- f Southern Oscillation

Models

RW
RW + NAOI
RW + SOI
RW + NAOI + SOI
DD
DD + NAOI
DD + SOI
DD + NAOI + SOI
DD + IDD
DD + IDD + NAOI
DD + IDD + SOI
DD + IDD + NAOI + SOI

Structural equation models



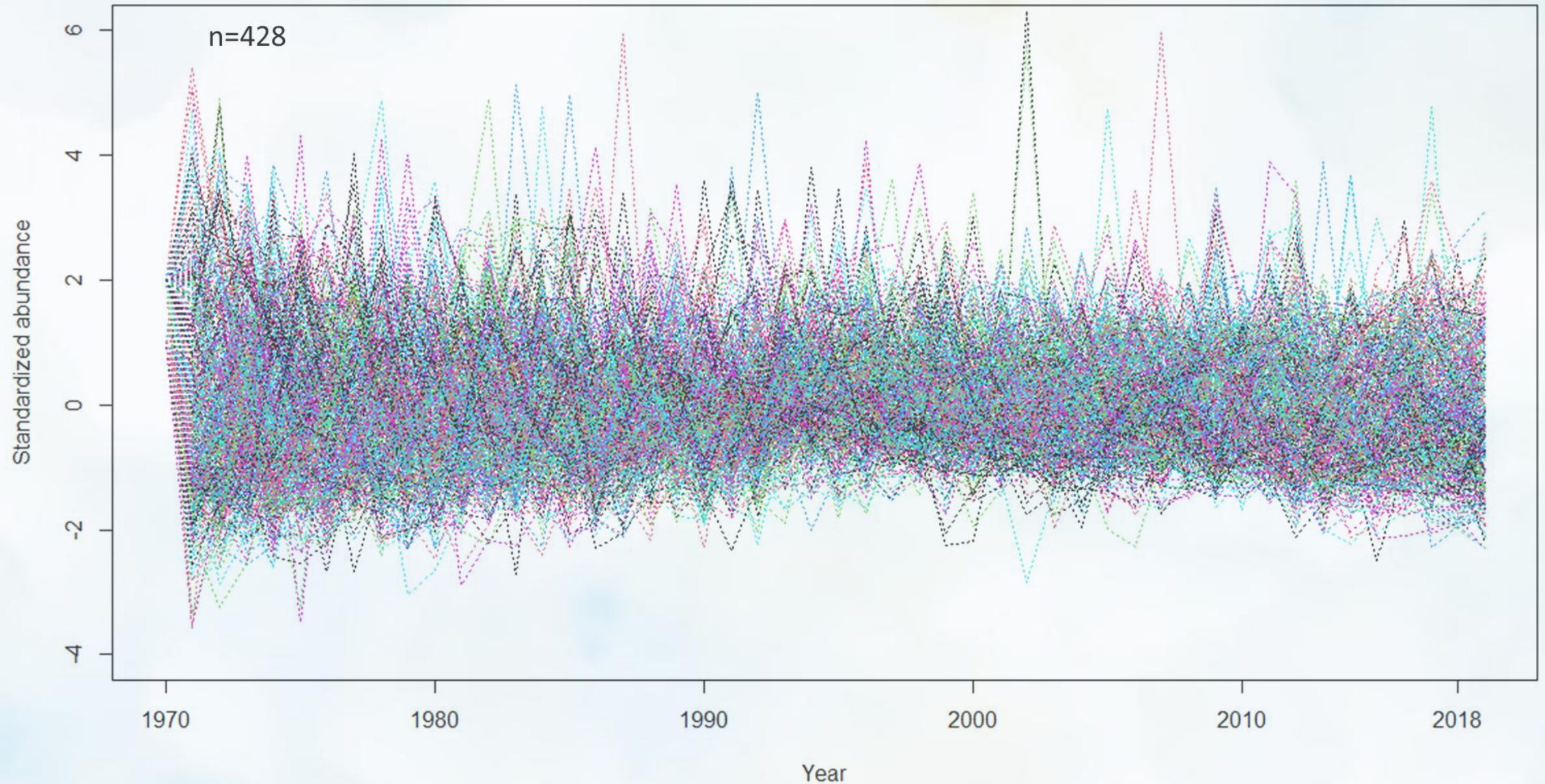


48 years (1970 – 2018) survey data

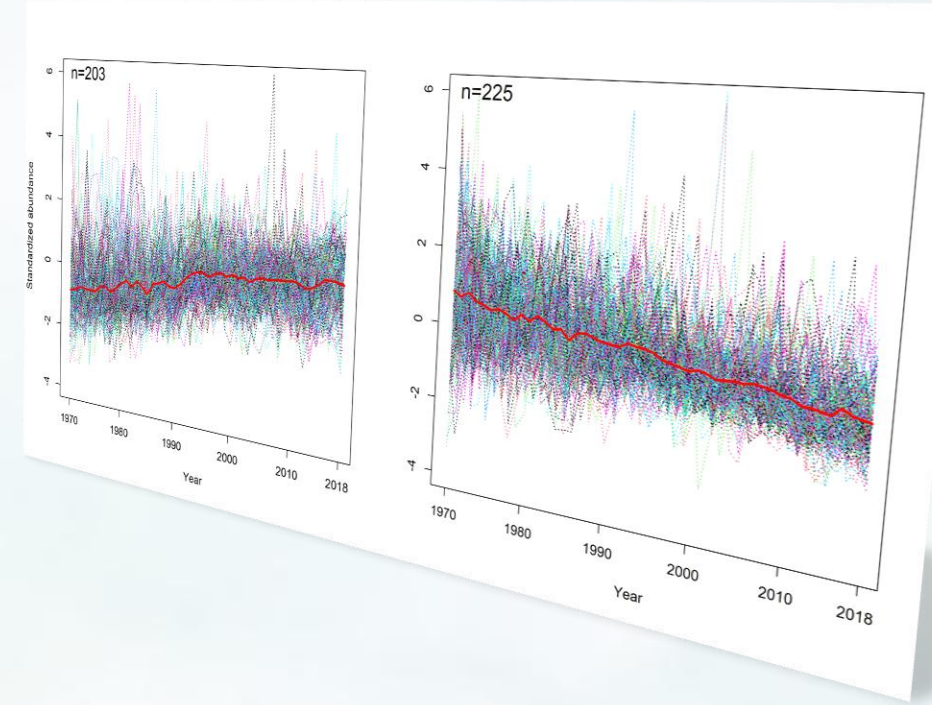
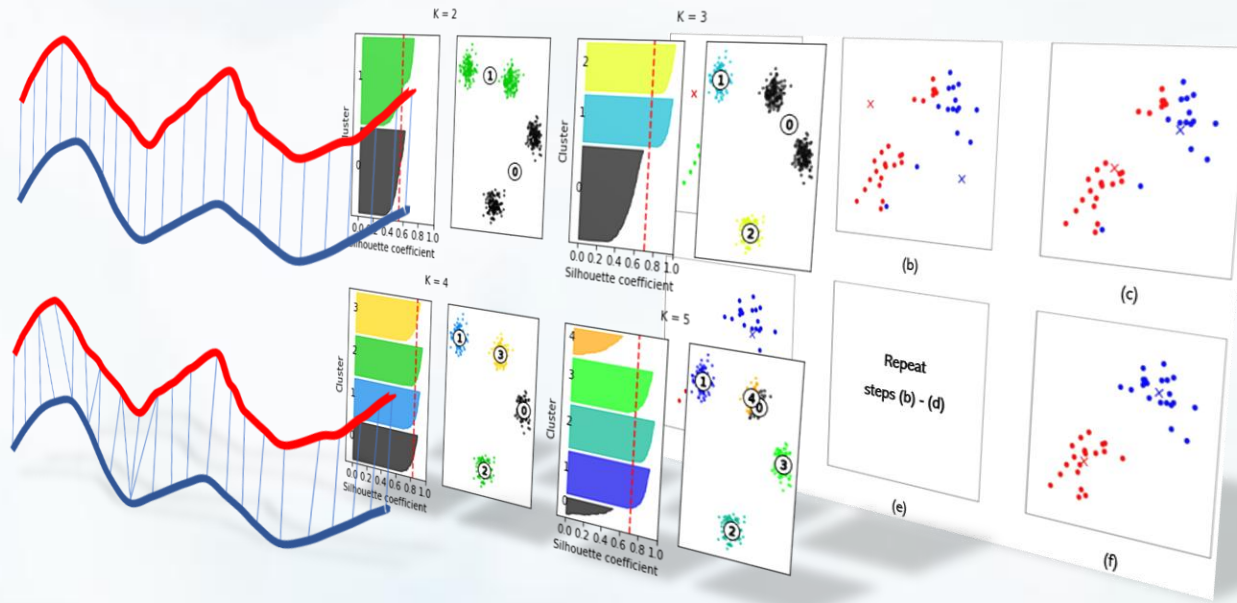
5092 routes covering **428** species

fit **5136** population models

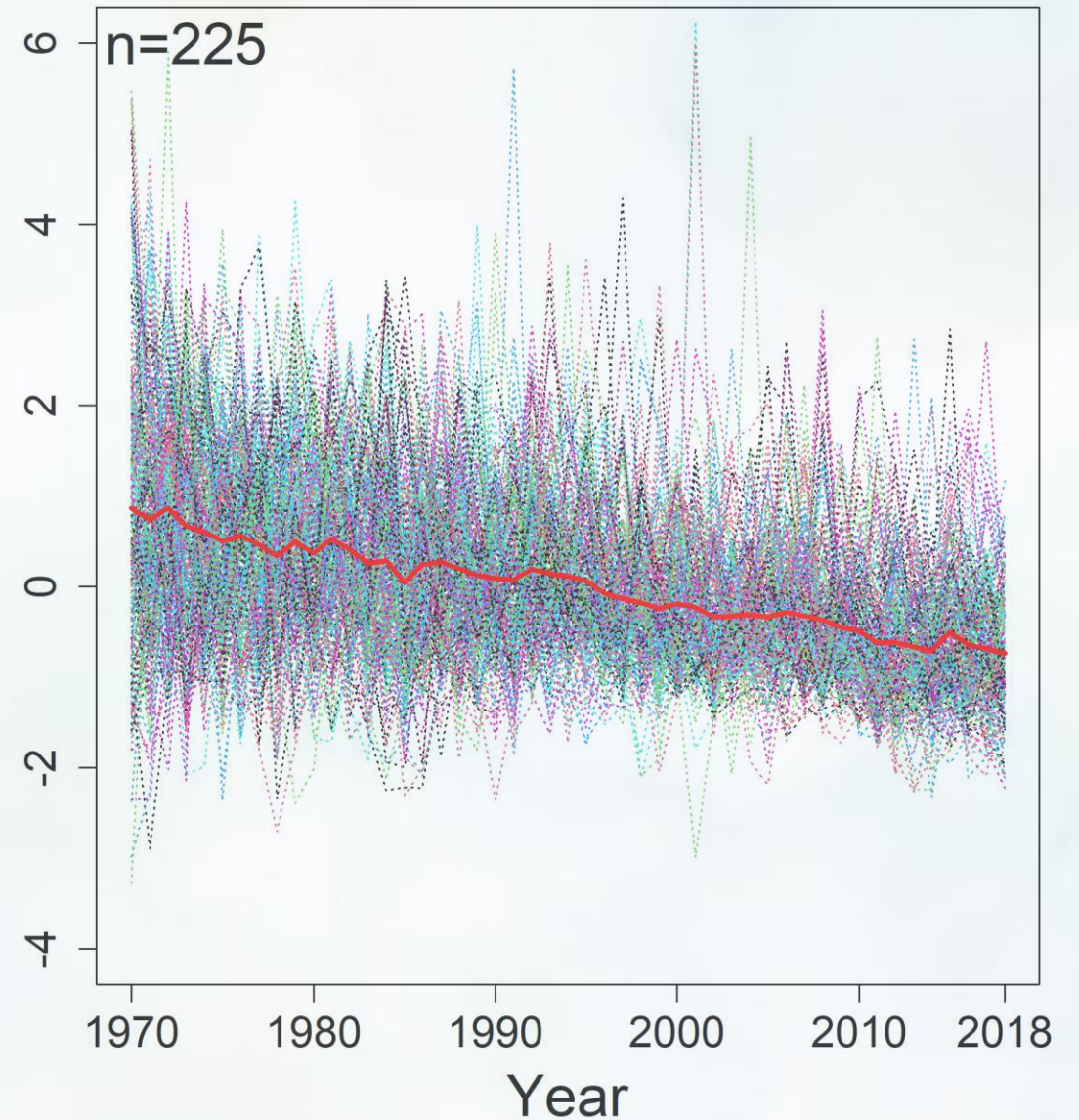
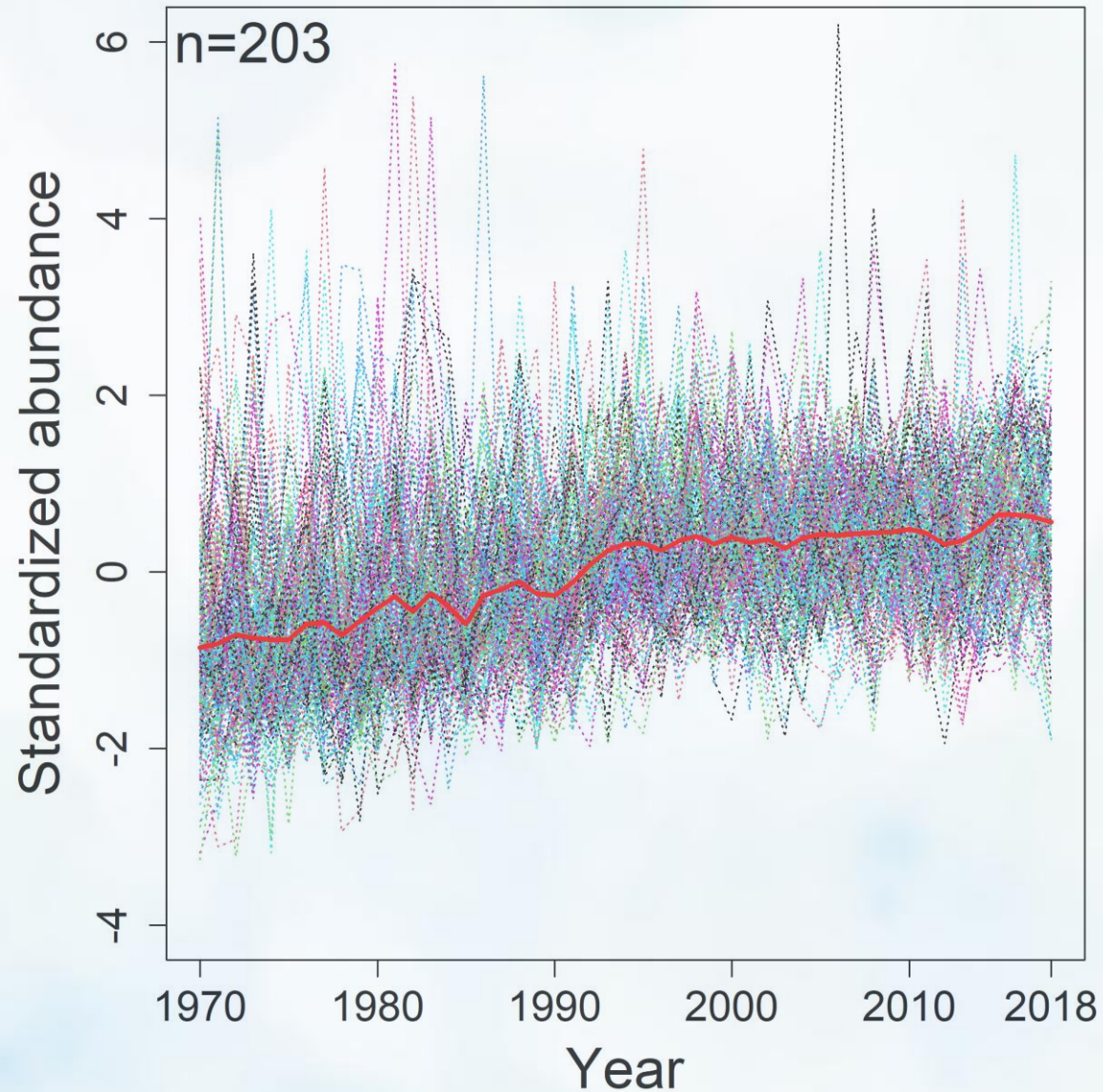
Time series of North American breeding birds

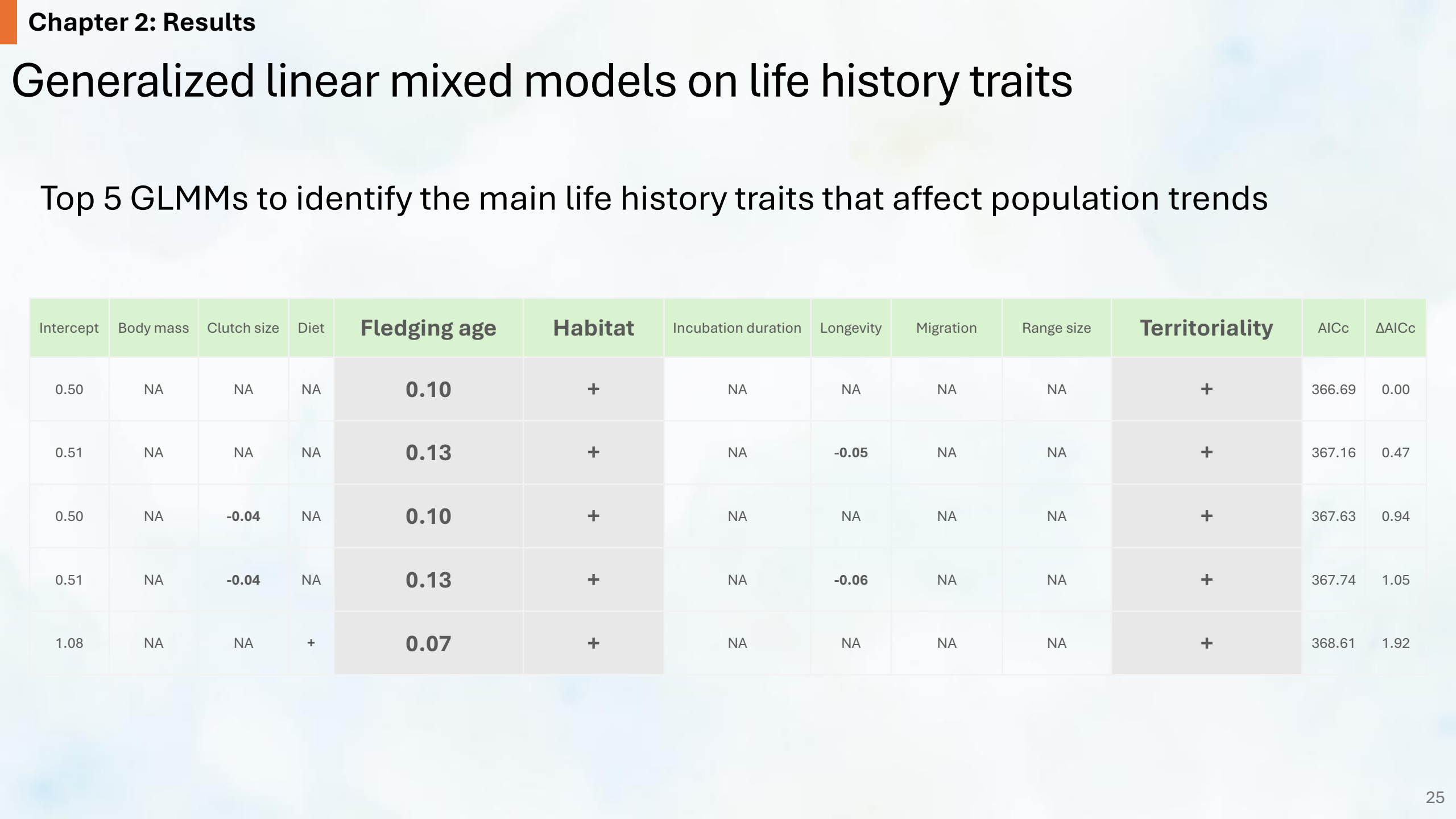


DTW K-means clustering



Temporal trends of North American breeding birds



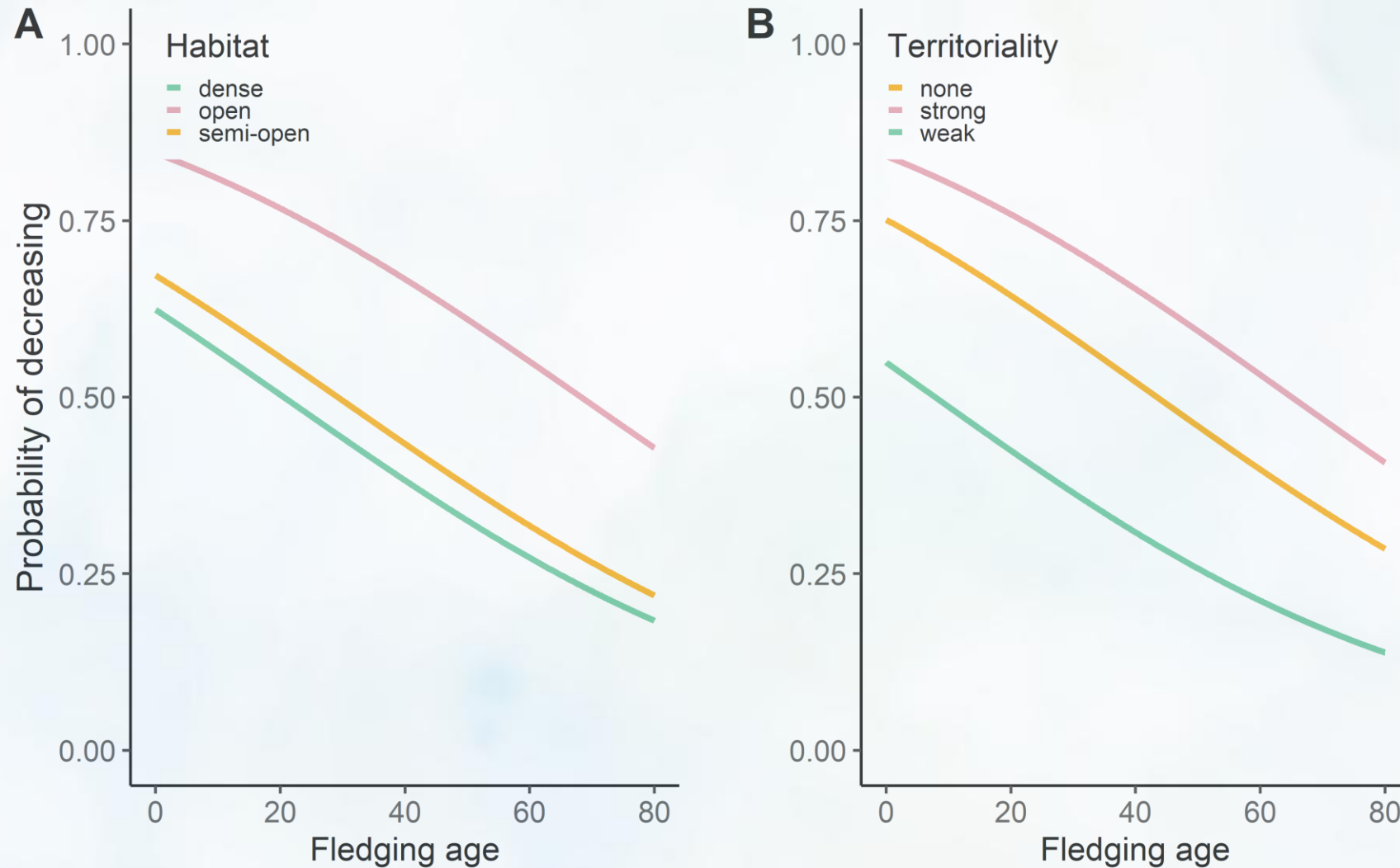


Bayesian phylogenetic logistic regression on life history traits

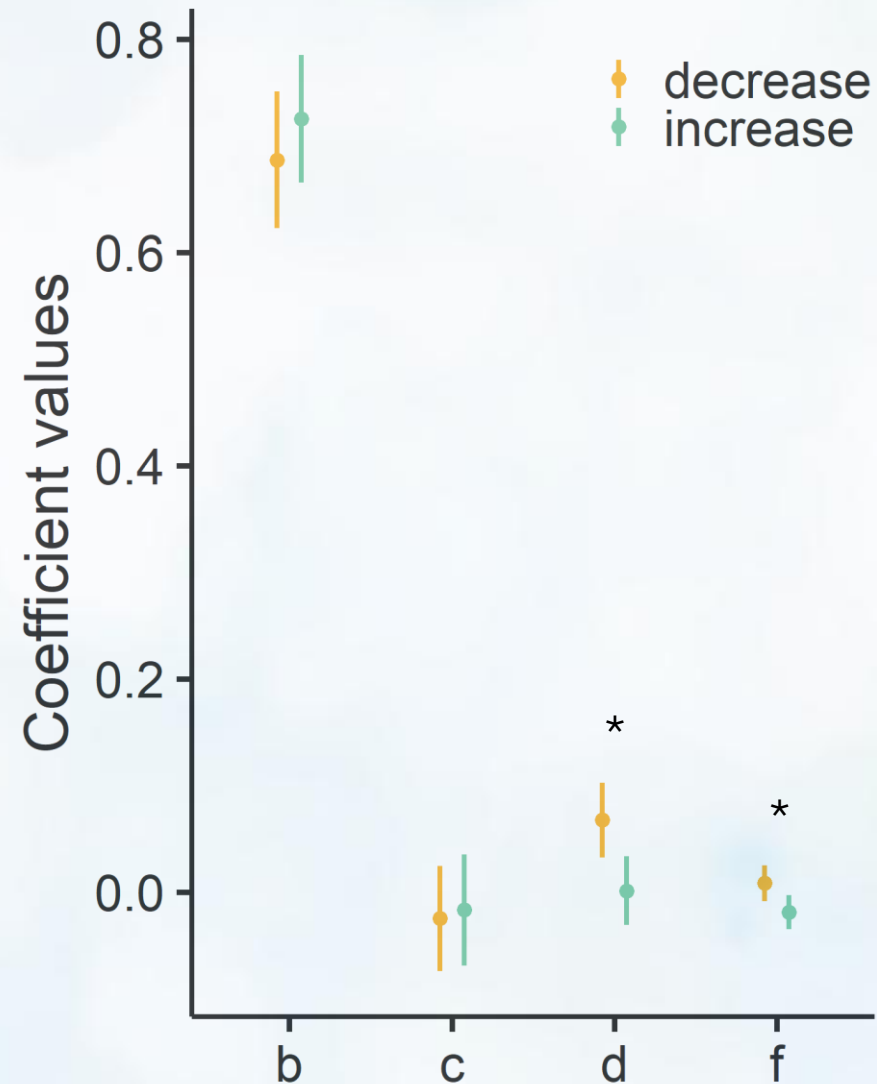
Variable	Coefficient	SE	LLM	ULM	$\hat{R}$
Intercept	1.3	1.72	-1.98	4.77	1.01
Clutch size	-0.12	0.25	-0.56	0.32	1.01
Fledging age*	0.59	0.33	0.01	1.23	1.01
Longevity	-0.14	0.26	-0.64	0.35	1.01
Territoriality-strong	-0.14	0.66	-1.39	1.04	1.01
Territoriality-weak*	1.34	0.48	0.47	2.26	1.01
Habitat-semi-open	0.01	0.45	-0.84	0.86	1
Habitat-open*	-1.31	0.55	-2.32	-0.36	1.01
Diet-invertebrates	-1.69	1.65	-5.04	1.44	1.01
Diet-nectar	-0.32	1.95	-4.19	3.47	1
Diet-omnivore	-2.24	1.67	-5.64	0.95	1
Diet-plants	-0.97	1.8	-4.56	2.5	1.01
Diet-scav	3.61	3.46	-2.31	11.21	1
Diet-seeds	-3.14	1.73	-6.65	0.16	1
Diet-vertebrates	-1.96	1.72	-5.44	1.3	1

Fledging age, habitat types, and territoriality significantly affected population trends

Bayesian phylogenetic logistic regression on life history traits



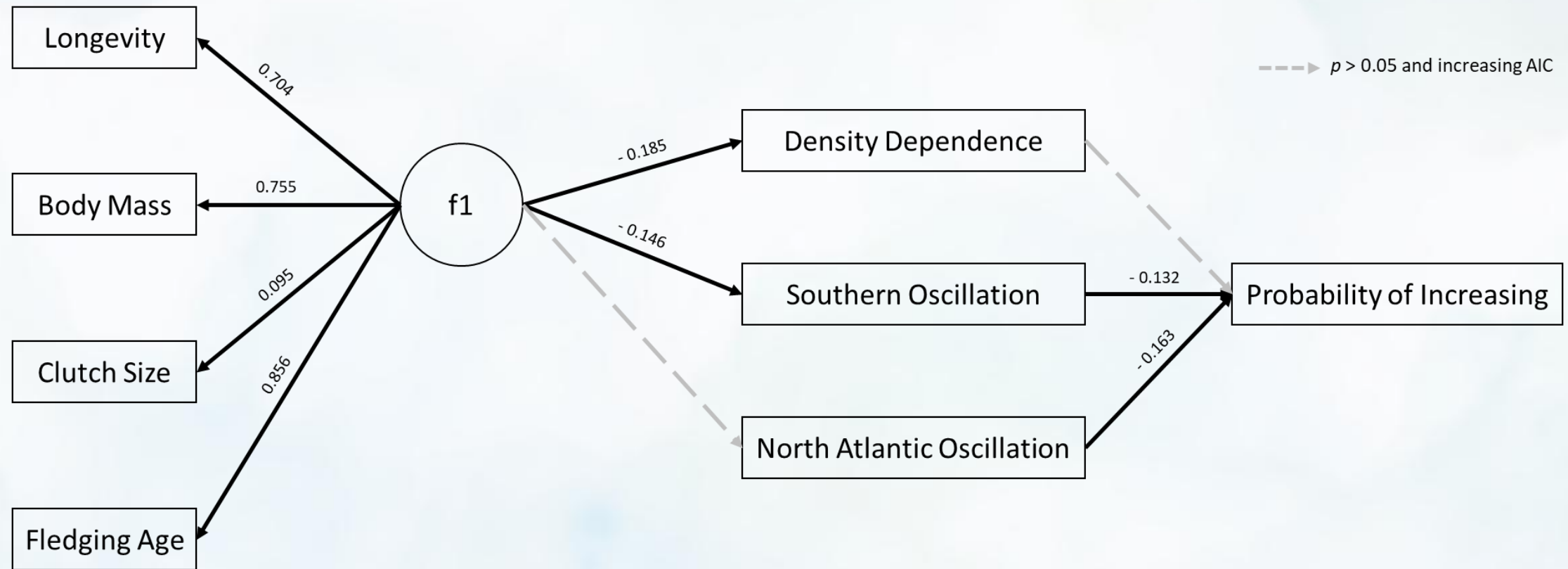
Population parameters of Gompertz SSM



The averages of the NAOI and SOI coefficients of increasing clusters were significantly less than those of decreasing cluster.

- b* density dependence
- c* delayed density dependence
- d* North Atlantic Oscillation
- f* Southern Oscillation

Structural equation models with nested random effect of phylogeny



Conclusions

Temporal patterns

DTW K-means clustering identified two distinct temporal dynamic trends in North American breeding bird populations.

Causal factors

Life history traits, density dependence, and responses to climate changes interact to determine these population trends. Specifically, life history traits indirectly affect population trends by mediating the effects of climate.

Chapter 3

Spatial variations of North American Breeding Bird Populations
within and between species

Objectives

1

Identify different patterns in the population trajectories of North American breeding birds for the past 50 years using machine learning;

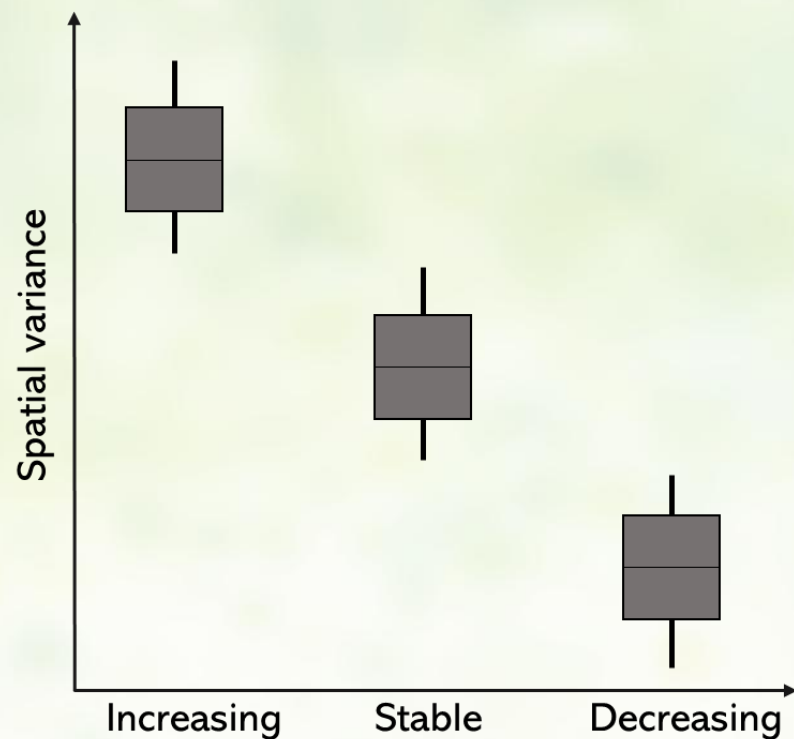
2

Explain the identified dynamic patterns using life history theory, phylogeny, population regulation, and **global** climate change;

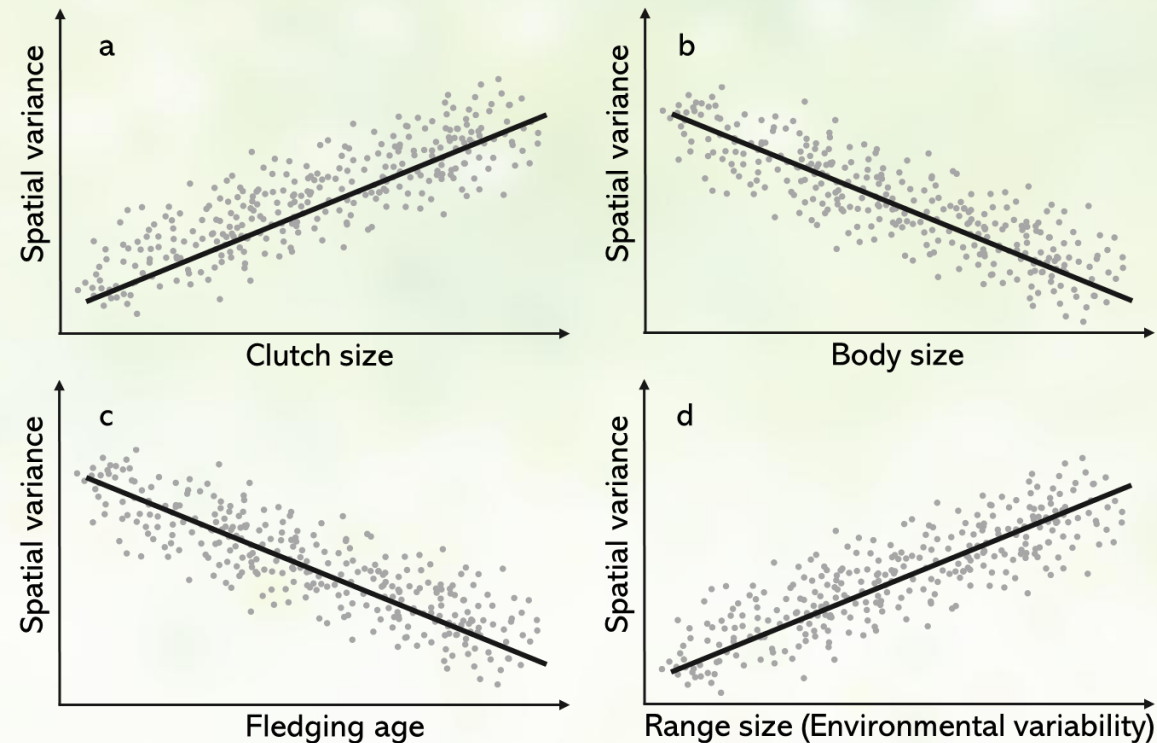
3

Investigate the spatial variations in avian population responses to **local** climate changes and avian abundances.

Hypotheses



Species with increasing or stable population trends would have greater spatial variability in the population responses to climatic changes than the species with decreasing population trends.

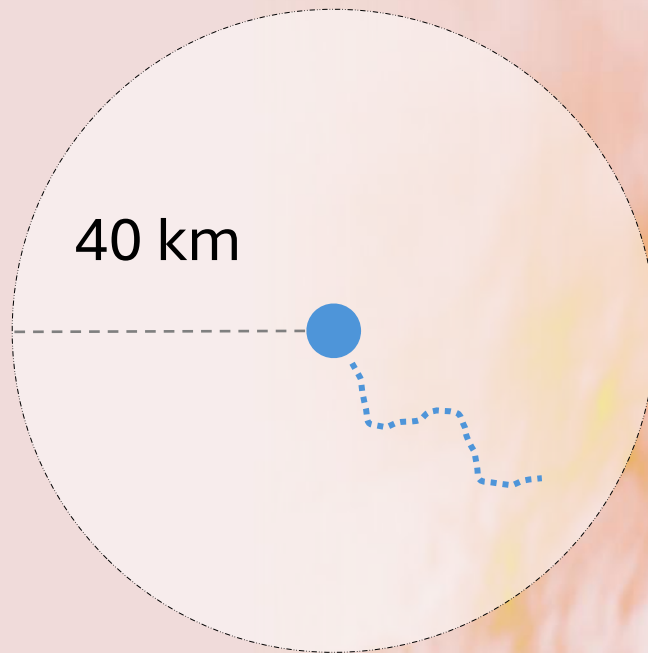


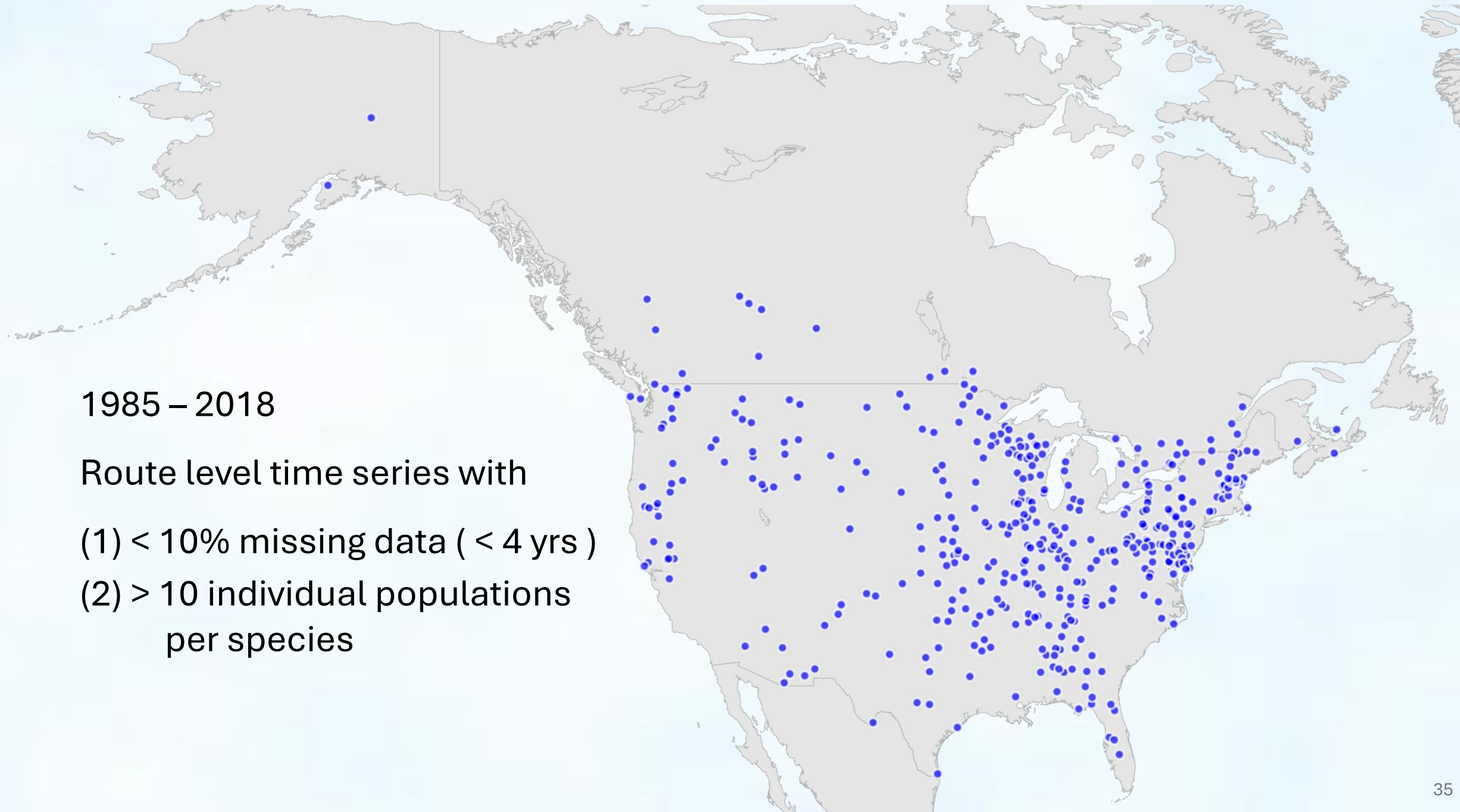
Life history strategies for faster population turnover or growth would result in more plasticity or spatial variation of population responses



Resolution: 30 arc sec (~ 1 km)

1983-2017





1985 – 2018

Route level time series with

(1) $< 10\%$ missing data (< 4 yrs)

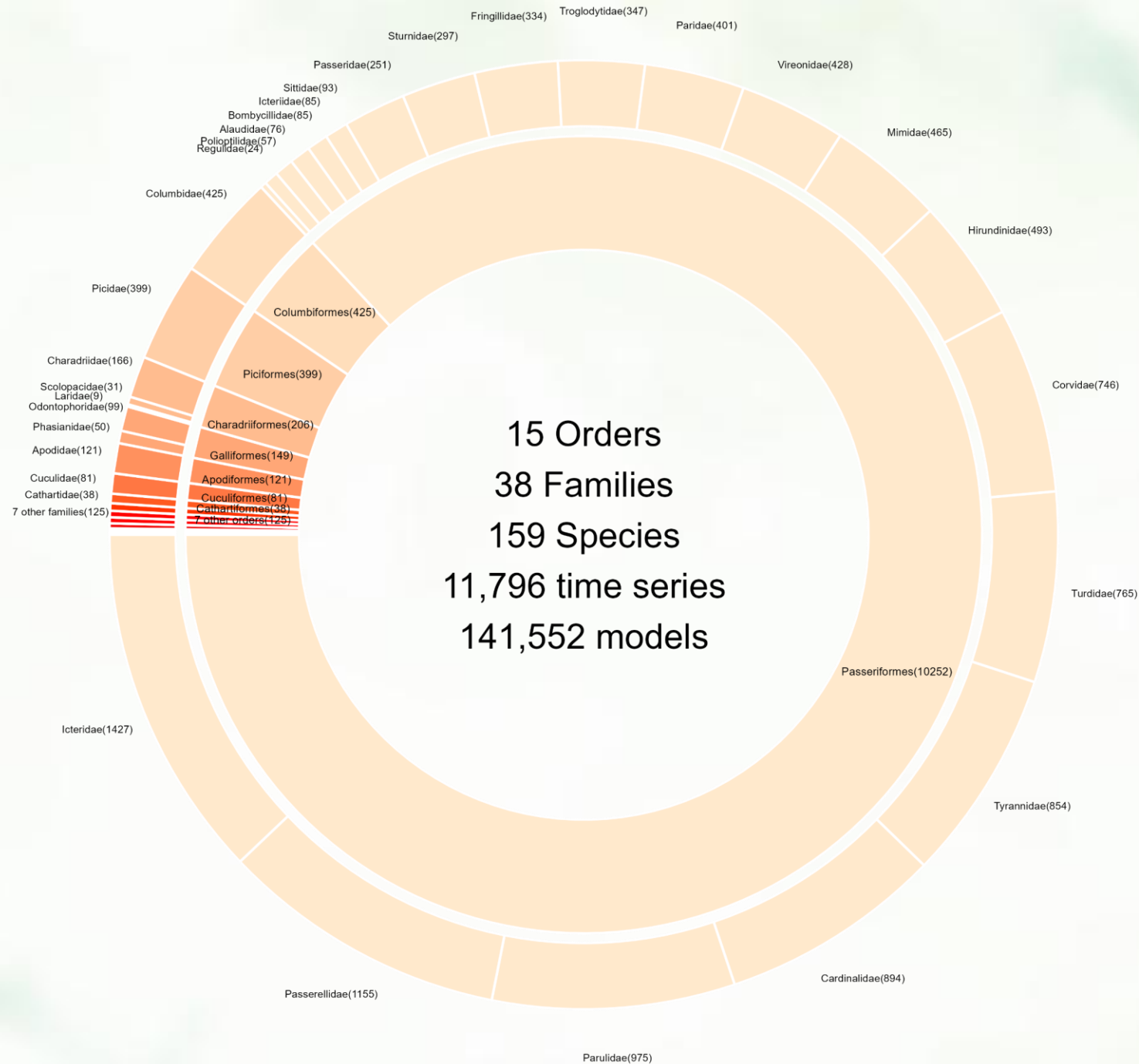
(2) > 10 individual populations
per species

Gompertz state space population growth models

$$Y_t = X_t + u_t$$

$$X_t = a + bX_{t-1} + cX_{t-2} + d\textit{Precipitation}_{t-1} + f\textit{Temperature}_{t-1} + e_t$$

- b density dependence
- c delayed density dependence
- d Precipitation
- f Temperature



Within species variations



n=250

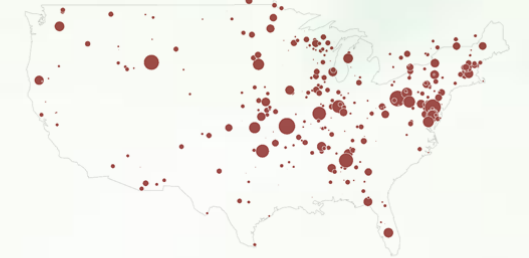
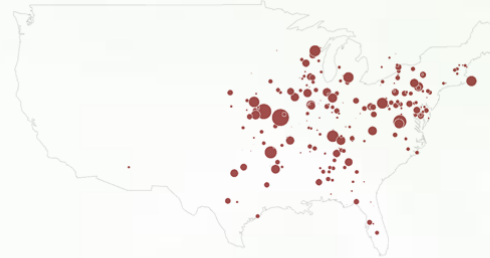


n=86



n=353

Density dependence



Delayed density dependence

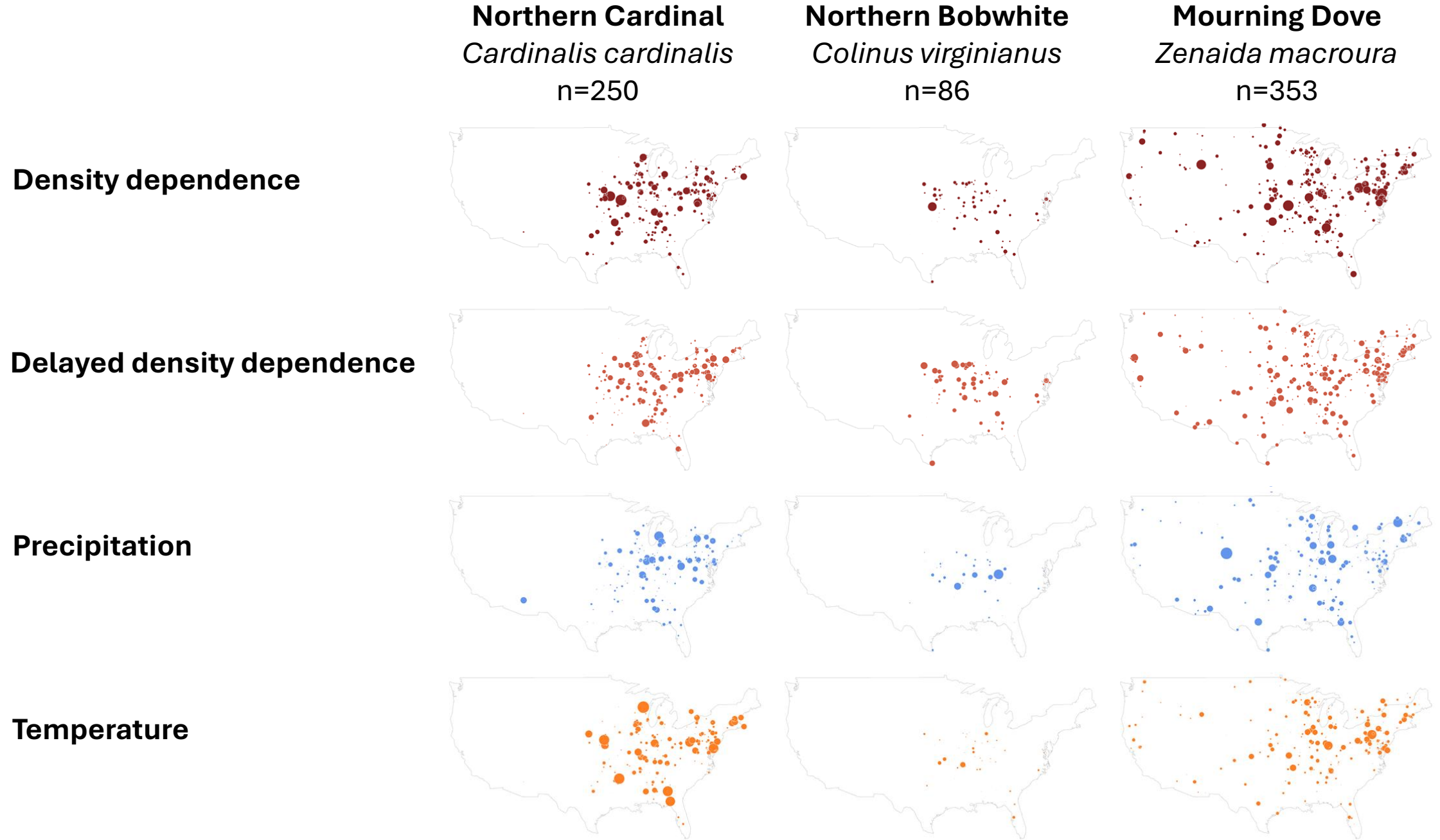


Precipitation

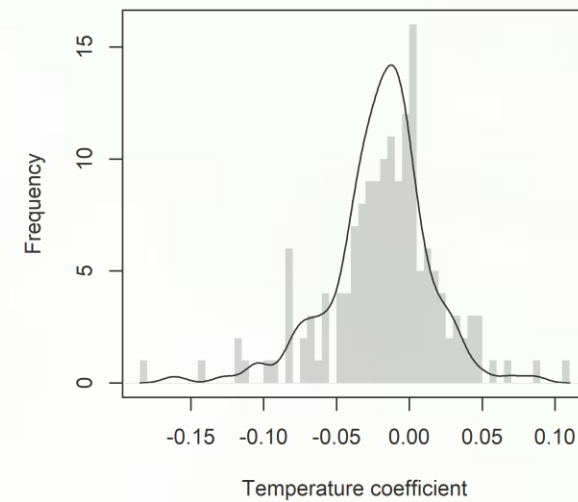
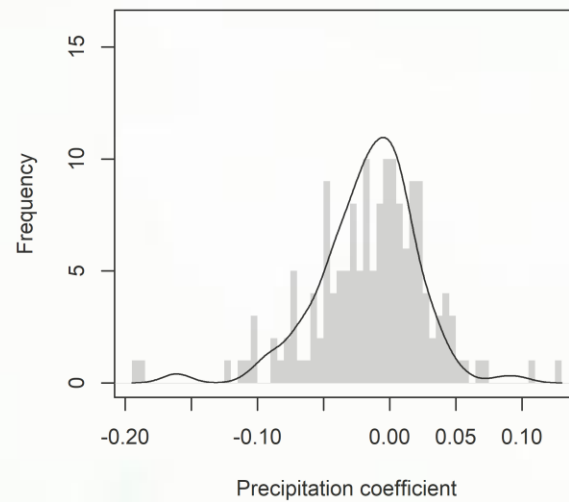
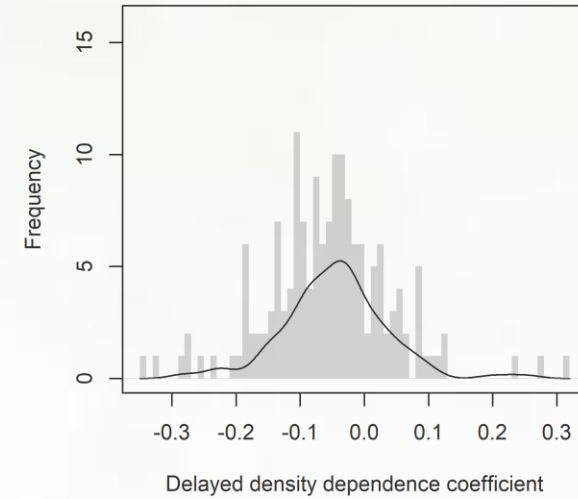
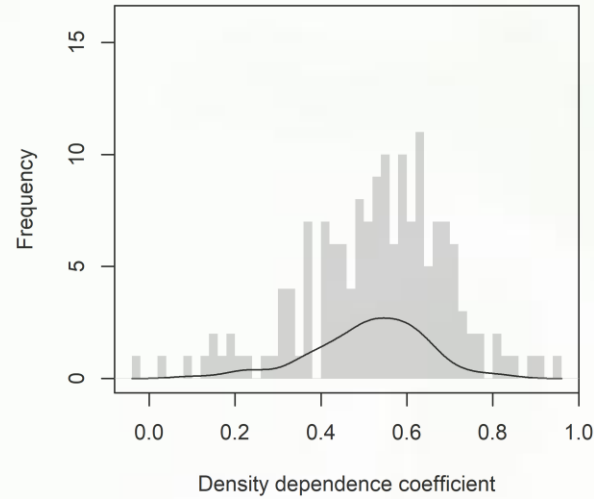


Temperature

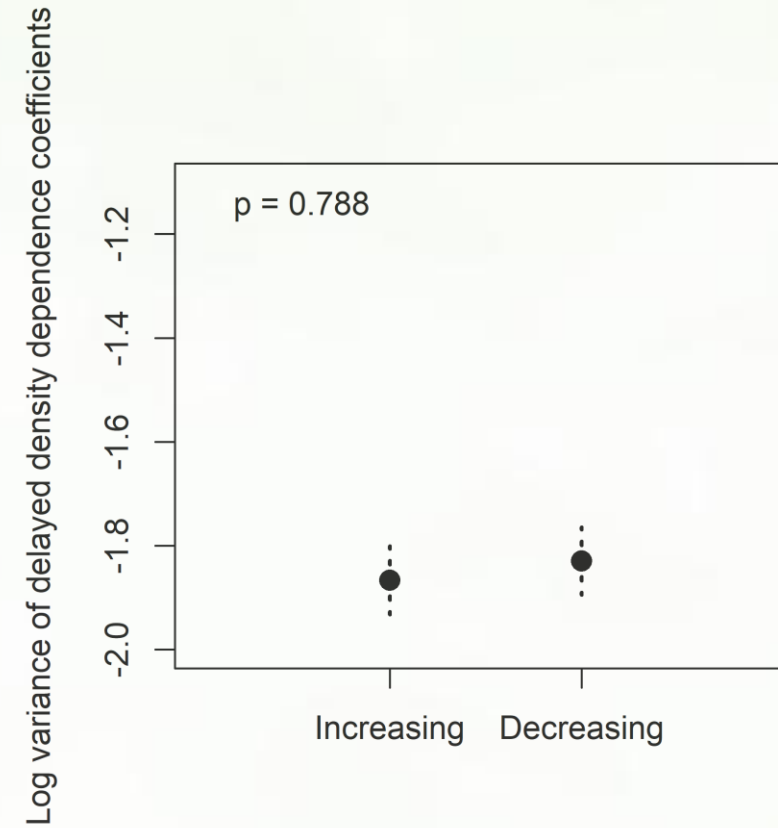
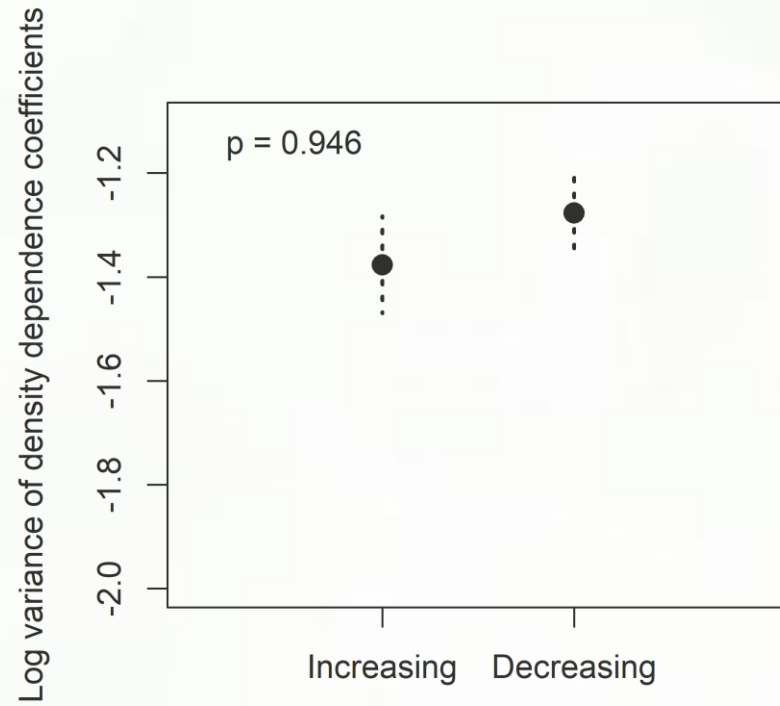




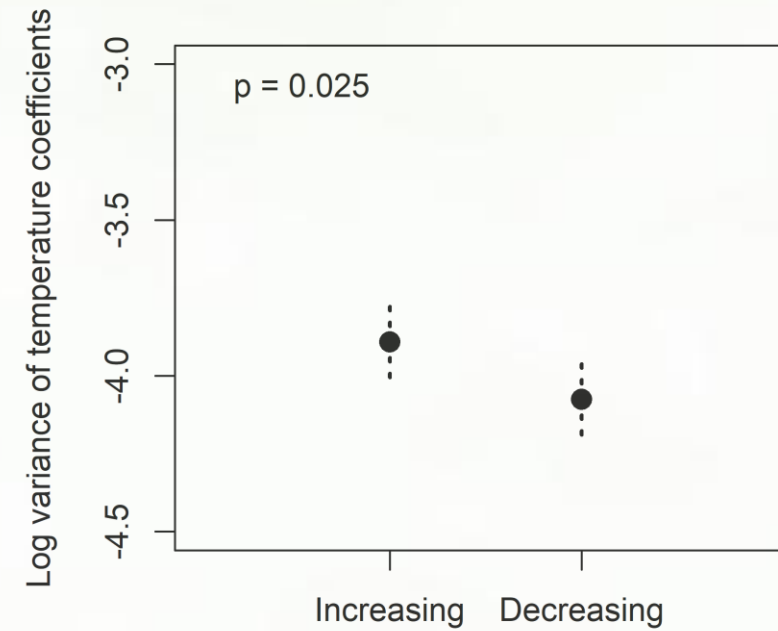
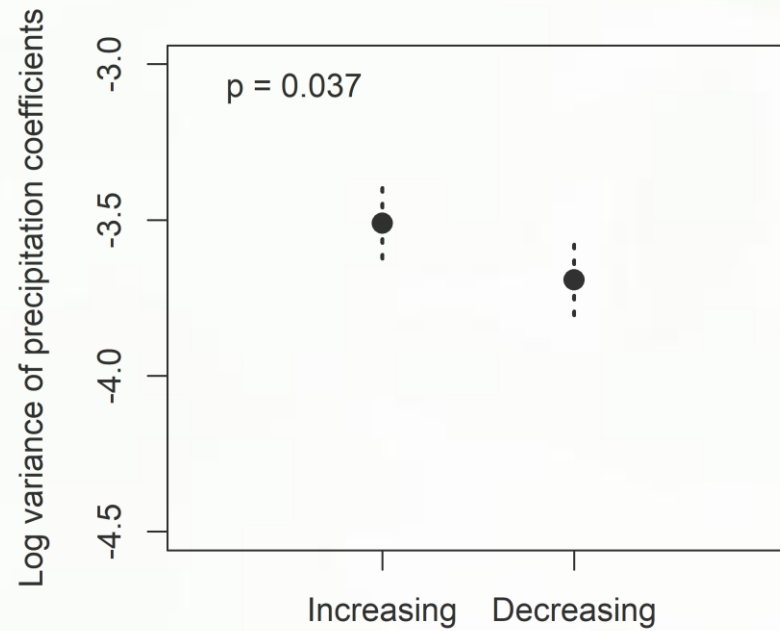
Empirical distribution of the means of population parameter estimates



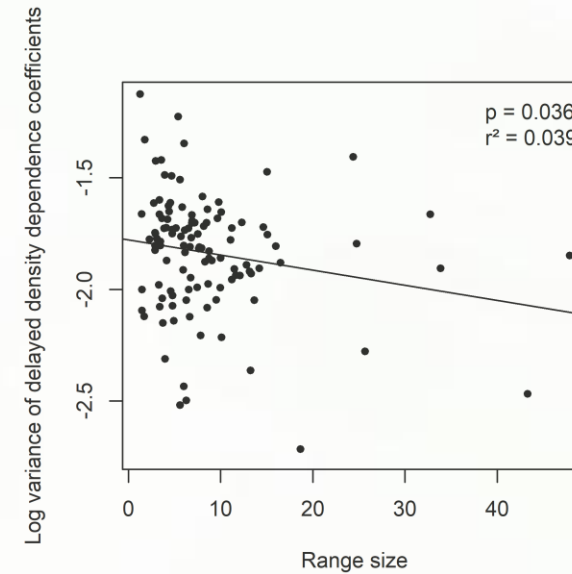
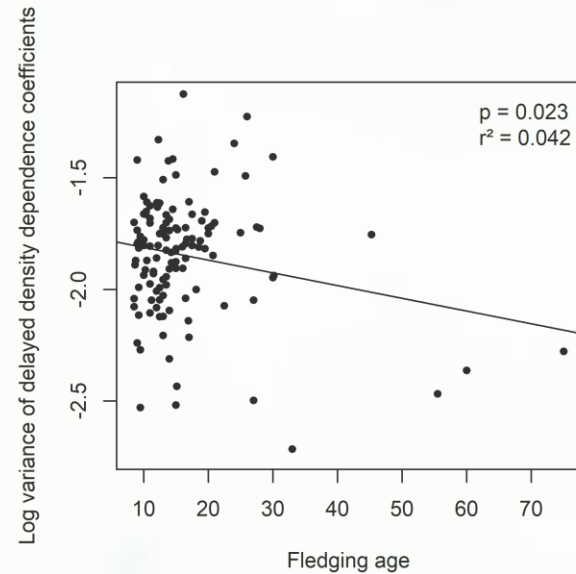
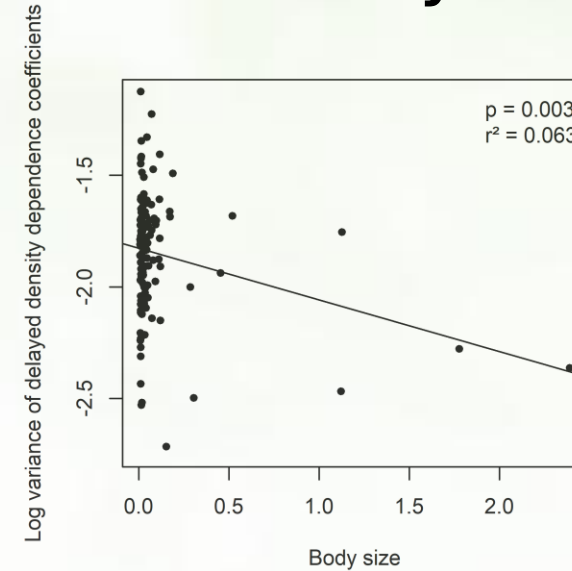
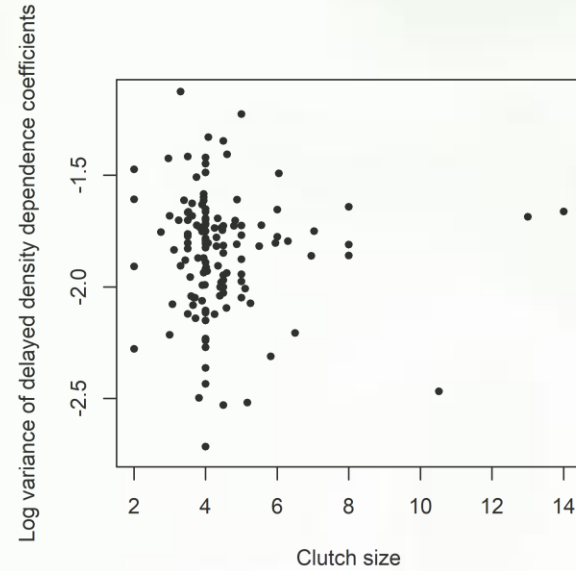
Comparisons of variances of population parameter estimates



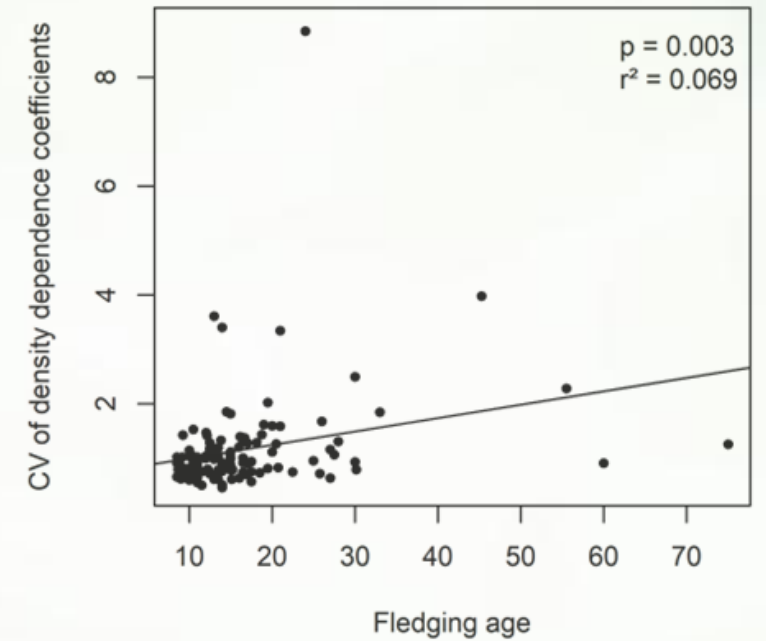
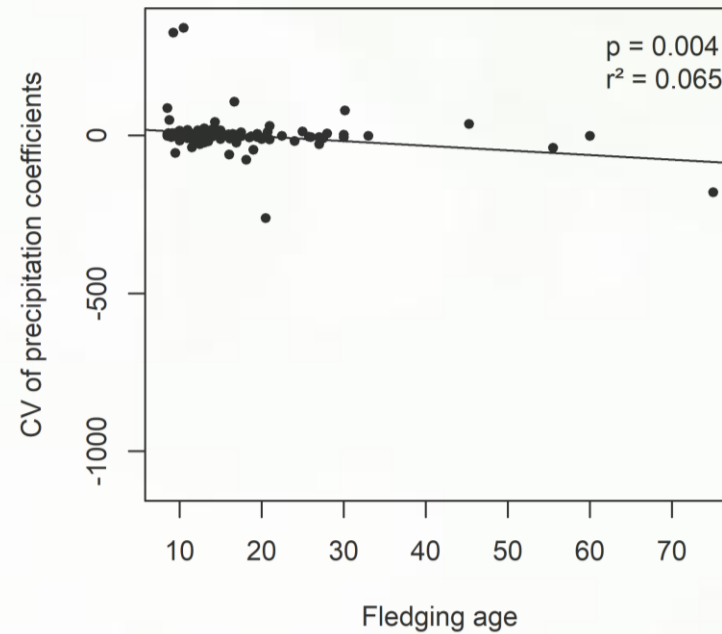
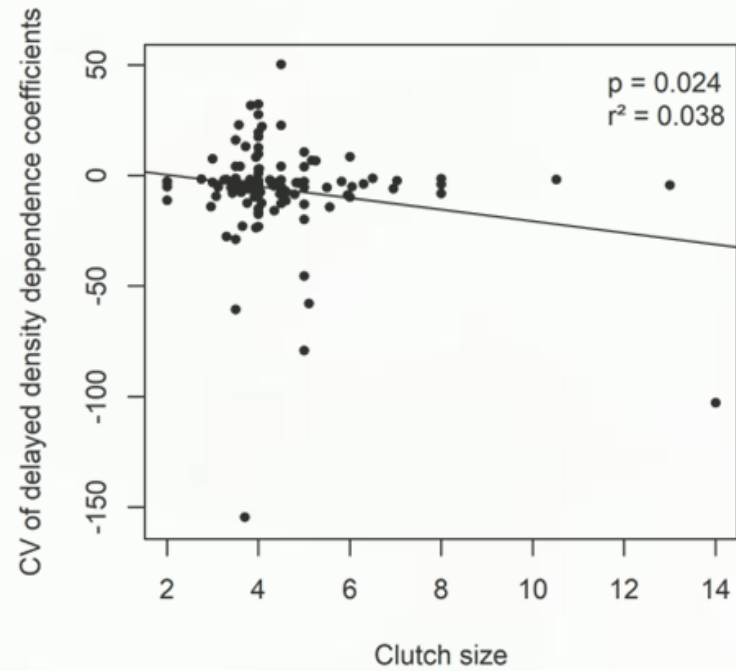
Comparisons of variances of population parameter estimates



Variances of parameter estimates on life history traits



CVs of parameter estimates on life history traits



Conclusions

Species with increasing trends showed greater spatial variations in their responses to climate than those with decreasing trends, consistent with the findings of Chapter 2. However, there were no significant differences in the spatial variation of the strength of population regulation between the two trend groups.

The relationships between life history traits and among-species spatial variation in population responses were rather complex. No consistent relationships were found between life history traits and spatial variations in population responses to density dependence or climate.

Take home messages

- 1 Two long-term trends of North American breeding bird populations.
- 2 Life history traits determined the strength of density dependence and birds' response to SOI, while indirectly affected population trends by mediating the effects of climate.
- 3 Species with increasing trends showed greater spatial variations in their responses to climate than those with decreasing trends.



Future work

Land use and land cover
More species coverage (eBird and CBC)
Interspecific interaction



Acknowledgements



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